Dynamic Policy Evaluation under Selective Labels: Evidence from Human and Algorithmic Decision-Making in Sequential Allocation

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Evaluating sequential decision-making is challenging when outcomes are observed only for chosen actions, creating a selective labels problem that is compounded in dynamic settings where decisions influence future information and learning. This paper studies how to compare human and algorithmic policies in such environments using observational data, focusing on settings where decisions must balance exploration and exploitation over time.

We apply this framework to biomedical innovation, modeling scientists' trial allocation decisions as a combinatorial contextual bandit with delayed feedback using data from over 700,000 experiments in the Protein Structure Initiative. Benchmarking human decisions against a neural upper confidence bound algorithm, we find that the algorithm achieves approximately 18% more successful project completions under comparable constraints. We find that human decision-makers systematically overestimate success probabilities and allocate effort too diffusely across projects, leading to inefficient exploration. Correcting prediction errors accounts for the majority of the performance gap, while improvements in exploration behavior provide additional gains. This paper contributes a general framework for evaluating dynamic policies under selective labels and provides large-scale empirical evidence that algorithmic policies can outperform human experts in complex sequential allocation tasks.

Key words: human-algorithm comparison; selective-label bias; dynamic decision-making; combinatorial bandits; structural estimation; exploration-exploitation

1. Introduction

Many important decisions in science, innovation, and operations are inherently dynamic. Decision-makers repeatedly allocate scarce resources across uncertain alternatives while learning from past outcomes. In these settings, decisions generate not only immediate outcomes but also information that shapes future choices, creating a fundamental trade-off between exploration and exploitation. A large literature has developed algorithmic approaches to such sequential decision problems, particularly within the multi-armed bandit framework and its extensions (Gittins and Jones 1979, Bubeck and Cesa-Bianchi 2012, Zhou et al. 2023). However, in many high-stakes domains—such as scientific research, clinical experimentation, and R&D portfolio management—these decisions continue to be made by human experts. This raises a natural question: can algorithmic policies improve decision-making in such environments?

Evaluating algorithmic policies in practice is challenging because the available data are generated by the decisions that were actually made. This creates a "selective labels" problem: we observe outcomes only for the actions selected by human decision-makers, while the outcomes of alternative actions—such as those an algorithm would have chosen—are unobserved (Lakkaraju et al. 2017). A natural approach is to evaluate an algorithm on the same set of cases for which outcomes are observed and compare its performance to that of human decisions on those cases. While this yields a well-defined comparison, it answers a limited question: it measures how the algorithm would perform on the cases selected by the human policy. Because this set of cases is endogenously determined by human decisions, it need not coincide with the cases the algorithm would choose if deployed in practice. As a result, this comparison does not recover the performance of the algorithm under its own decision rule.

In static, one-shot settings, this limitation can be addressed using contraction-based approaches that exploit variation across decision-makers (Kleinberg et al. 2018, Fu et al. 2021). For example, in judicial bail decisions, more lenient judges release a larger set of defendants, including some whom stricter judges would detain. Because cases are quasi-randomly assigned to judges, outcomes are observed for this expanded set. To compare policies at a fixed release rate, one can take the set of defendants released by a lenient judge, apply the algorithm to rank those defendants, and select a subset equal in size to the release rate of a stricter judge. The outcomes of these selected defendants are then compared to the outcomes of the defendants actually released by the stricter judge. This procedure allows both policies to be evaluated at the same decision threshold using only observed outcomes, without requiring a model of counterfactual behavior.

In dynamic settings such as ours, however, this type of variation is not available. We observe decisions made under a single realized policy and set of constraints, without comparable observations of how similar decision-makers would act under different conditions on a similar set of cases.

In the absence of such variation, a direct comparison between human and algorithmic policies cannot be constructed from the data alone.

To address this limitation, we restrict attention to a single decision-maker and to the set of actions for which outcomes are observed. Within this setting, an algorithmic policy can be directly constructed and applied to determine how decisions would be reallocated under alternative constraints, such as a reduced capacity. In contrast, the corresponding human behavior under these constraints is not observed: the data record only the decisions made under the realized conditions. To construct this counterfactual, we estimate a structural model of human decision-making using the observed data and use it to simulate how decisions would adjust under the same constraints. This enables both the algorithm and the human policy to be evaluated under comparable conditions on a common outcome space. We refer to this procedure as a *dynamic contraction* approach, as it extends contraction-based evaluation to dynamic settings in which policy variation must be constructed rather than observed.

We apply the dynamic contraction approach in the context of biomedical innovation, using data from the Protein Structure Initiative (PSI), a large-scale, NIH-funded scientific program spanning 16 years and involving over 700,000 experimental trials conducted by dozens of laboratories. In this setting, scientists repeatedly allocated scarce experimental capacity across a large set of protein structure projects under substantial uncertainty. Each project required multiple costly and time-consuming trials to determine a successful structure, with individual experiments costing approximately \$1,800 [1] and taking several months to complete. At each point in time, laboratories had to decide which projects to initiate or continue and how to allocate trials across them, subject to budget and operational constraints. These decisions involved a dynamic trade-off between exploiting projects with a higher perceived likelihood of success and exploring new or uncertain opportunities to generate information. The setting therefore provides a large-scale, real-world instance of dynamic resource allocation with delayed feedback, heterogeneous alternatives, and path-dependent learning.

We model the laboratory decision problem as a combinatorial contextual bandit, in which decision-makers allocate a limited number of experimental trials across a large set of projects in each period. Each project is characterized by observable features, and outcomes are realized with delay, so that learning unfolds over overlapping decision periods. We adapt the Combinatorial Neural Upper Confidence Bound (CN-UCB) algorithm to this setting. The algorithm uses project-level feature representations derived from biological and physicochemical attributes, incorporates delayed feedback, and accounts for the interdependence between past learning and future

[1] This is calculated by dividing the total funding for the PSI (\$1.269 billion) by the total number of trials attempted.

allocations. This allows it to balance exploration and exploitation in a high-dimensional, capacity-constrained environment using only information available at the time decisions are made.

We compare the performance of this algorithm to the decisions made by scientists during the production phase of the PSI (PSI-2, July 2005–June 2010), when laboratories were primarily tasked with maximizing the number of successfully determined protein structures. To enable a valid comparison in the presence of dynamic selective labels, we apply the dynamic contraction approach to construct comparable policy paths. We build and estimate a structural model of scientist trial allocation behavior. Using the estimated structural model, we simulate how scientists would have reallocated trials under alternative capacity constraints, generating contracted human policies that select subsets of actions observed in the data over time. This ensures that outcomes used for evaluation are empirically observed while preserving the temporal dependencies induced by sequential decision-making. We then evaluate both human and algorithmic policies on this common outcome space.

Our analysis reveals a substantial performance difference between human and algorithmic decision-making. For example, under 70% of original trial budgets, the CN-UCB algorithm achieves an 18.7% increase in successful project completions relative to human scientists. This advantage persists across a range of capacity levels and is robust to variation in project composition and observed outcomes. These findings indicate that, even in a complex, high-skill scientific environment with delayed feedback and domain expertise, algorithmic policies can systematically outperform human decision-making in dynamic allocation tasks.

To understand the source of this performance gap, we use the structural model to decompose differences in outcomes into two components: belief formation and exploration behavior. We find that scientists tend to overestimate project success probabilities and allocate trials across too many projects, resulting in shallow investment and frequent abandonment. Counterfactual simulations show that correcting prediction errors alone recovers approximately 72% of the performance gap, while correcting exploration behavior alone recovers about 34%. Only when both components are adjusted jointly does performance approach that of the algorithmic benchmark. These results highlight the interaction between beliefs and allocation decisions in shaping dynamic performance and suggest that inefficiencies arise from both miscalibrated expectations and suboptimal learning strategies.

This paper makes two contributions. First, we develop a method for evaluating decision policies in dynamic environments with selective labels. Existing approaches to selective labels rely on variation across decision-makers to construct valid comparisons using observed outcomes, and are therefore limited to static, one-shot settings. In contrast, dynamic environments typically lack such

variation: we observe decisions under a single realized policy and constraint, without comparable observations under alternative conditions. Our dynamic contraction approach addresses this limitation by constructing the missing policy variation while continuing to rely only on observed outcomes. By separating decisions from outcomes, we use a structural model to generate counterfactual decision paths under alternative constraints, while restricting evaluation to actions for which outcomes are observed in the data. This enables symmetric comparisons between human and algorithmic policies in settings where standard contraction-based approaches cannot be applied.

Second, we provide large-scale empirical evidence on human versus algorithmic decision-making in a complex, high-skill dynamic environment. Using data from a large scientific research program, we show that an algorithmic policy can substantially outperform human experts under comparable constraints. Importantly, our framework allows us to move beyond documenting performance differences to diagnosing their sources. We show that the performance gap arises from systematic differences in both belief formation and exploration behavior, with human decision-makers exhibiting overoptimistic beliefs and inefficient allocation of effort across projects. These findings highlight that the value of algorithmic decision support in dynamic environments stems not only from improved prediction, but also from more effective management of the exploration–exploitation trade-off.

The rest of this paper is organized as follows: Section 2 reviews related literature. Section 3 introduces our research context and data. Section 4 describes the ML algorithm and human decision making models. Section 5 presents the comparison of machine and human decisions. Section 6 diagnoses the inefficiencies in human decision making. Section 7 concludes and discusses potential next steps.

2. Relevant Literature

A growing literature compares human and algorithmic decision-making across a range of domains, including judicial bail decisions (Kleinberg et al. 2018, Rambachan 2024), lending (Fu et al. 2021, Jansen et al. 2025), news curation (Peukert et al. 2024) and medical diagnosis (Ribers and Ullrich 2024, Mullainathan and Obermeyer 2022). These studies typically focus on static, one-shot settings, where each decision is made once and outcomes are observed for a subset of actions. In such environments, researchers have documented systematic differences between human judgment and algorithmic predictions, often finding that algorithms can improve decision quality. Relatively little work examines this comparison in dynamic settings, where decisions are made sequentially and learning plays a central role. An exception is recent work that considers hiring as a contextual bandit problem where algorithms must balance hiring from proven groups (exploitation) with hiring from under-represented groups (exploration) to discover new talent (Li et al. 2025). However, this

line of work abstracts from the selective labels problem and therefore does not address the challenge of evaluating algorithmic policies using observational data.

Another stream of literature addresses the selective labels problem that arises when outcomes are observed only for actions that are taken (Lakkaraju et al. 2017). In static settings, several approaches have been proposed to enable policy evaluation without imputing unobserved outcomes. Contraction-based methods exploit variation across decision-makers: when cases are quasi-randomly assigned to individuals with different decision thresholds, outcomes are observed for a broader set of cases, allowing policies to be compared at a common decision rate using only observed outcomes (Kleinberg et al. 2018). The validity of this approach relies on the presence of such variation and on the comparability of cases across decision-makers.

An alternative approach uses extrapolation based on quasi-random assignment, inferring counterfactual outcomes by comparing decisions and outcomes across decision-makers with different propensities to act (Angelova et al. 2025). This approach requires strong assumptions on the stability of outcome distributions across decision-makers and sufficient overlap in the support of decisions. While these methods provide credible comparisons in one-shot environments, they rely on cross-sectional variation and separability of decisions, assumptions that do not extend to dynamic settings where decisions affect future information and the data generated by different policies diverge over time. Our paper contributes to these literatures by developing a method to evaluate and compare human and algorithmic decision-making in dynamic environments with selective labels.

Our work also relates to off-policy evaluation (OPE), which estimates a target policy's value using data generated by another policy. In contextual bandits, inverse propensity weighting adjusts observed rewards for differences in action-selection probabilities, direct methods estimate rewards, and doubly robust estimators combine these approaches (Dudík et al. 2014). These ideas extend beyond static decisions. Jiang and Li (2016) develop sequential doubly robust evaluation, while Kallus and Uehara (2020, 2022) combine value-function estimation with marginalized density ratios to improve efficiency under Markov structure. Alternatively, fitted Q evaluation estimates a target policy's value through recursive regression without requiring propensity weights or an explicit transition model (Le et al. 2019).

These methods provide important tools for dynamic evaluation, but their applicability depends on the available support and the credibility of the required statistical models. Weighting cannot recover the contribution of actions with zero probability under the historical policy; evaluating such actions through reward or value models requires additional extrapolation assumptions (Sachdeva et al. 2020). Our setting involves joint trial allocations across many projects and information histories that change with past allocations. Observing a project-trial pair somewhere in

the historical record does not establish support for selecting it under every counterfactual information history. Applying conventional estimators would therefore require additional justification for history-conditional overlap, estimation of the relevant propensities or density ratios, or generalization of outcome or value functions. Related work explicitly recognizes these limitations: Sachdeva et al. (2020) study action- and policy-space restrictions under deficient support, and Fujimoto et al. (2019) constrain offline reinforcement learning to reduce extrapolation error.

A closely related alternative is replay-based evaluation. Li et al. (2011) evaluate adaptive contextual-bandit algorithms by retaining logged events whose actions match the algorithm's choices. Under uniformly randomized logging and their sampling assumptions, replay provides unbiased evaluation without outcome imputation or explicit propensity reweighting. Our historical scientific allocations do not arise from such a randomized logging design. We instead construct a capacity-based diagnostic comparison conditional on the empirically observed outcome space.

3. Research Context and Data Description

We demonstrate our study framework in a biomedical innovation context, specifically, the Protein Structure Initiative (PSI), funded $1.3 billion by the National Institutes of Health (NIH) from 2000 to 2015.

3.1. Research Context

The goal of PSI is to determine the 3-D structures of a set of protein molecules; see Figure 1 for an example. The determination of the 3-D structures of protein molecules is critical for drug development (Van Montfort and Workman 2017) and vaccine development (Wrapp et al. 2020).[2] While the scientific context is specialized, the underlying decision problem is general: scientists must sequentially allocate costly experimental effort across a large portfolio of uncertain projects, under capacity constraints, with delayed feedback and cross-project learning.

In this setting, a "project" is the determination of the 3-D structure of a specific protein sequence, which in turn serves as a unique identifier of the project. Scientists allocate discrete units of experimental effort, called "trials," to each project. These trials are parallel, independent attempts to determine the protein structure.[3] A project is considered successful if at least one of its trials is successful, thereby determining the protein's 3-D structure.

Objectives. The PSI was organized into three phases (NIGMS 2011): a pilot phase (PSI-1, 2000–2005) that developed experimental pipelines, a production phase (PSI-2, 2005–2010)

[2] Several Nobel Prizes have been awarded to researchers in this area, and the most recent one was the 2024 Nobel Prize in Chemistry for AlphaFold.

[3] Each trial involves multiple sequential stages: preparing the DNA sequence to produce the target protein, producing and purifying the protein, inducing crystals from the protein, and finally examining the protein structure using technologies like x-ray diffraction.

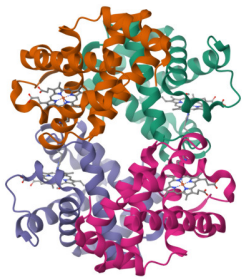


Figure 1 Example 3-D structure of a protein molecule
Source:
https://www.rcsb.org/structure/9VIB.

that aimed to maximize the number of structures determined, and a biomedical phase (PSI-3, 2010–2015) that prioritized projects of higher biological importance. We focus on PSI-2, where the primary goal was to maximize the total number of unique protein structures determined (i.e., unique successful projects) (Dessailly et al. 2009). This clarity of objective enables a well-defined optimization problem. We show robustness to alternative objectives (such as project novelty, biomedical importance, and organism specificity of protein sequences) in Section 6.

Decisions. In each period, participating labs select which *projects* to pursue and decide how many *trials* to allocate to each pursued project, subject to resource capacity constraints. These choices are highly complex, uncertain, and costly. First, labs have a large choice set of potential projects to choose from but most are highly risky—on average, fewer than 3% of trials resulted in success. Success probabilities can vary widely across projects: some succeed after a single trial, while others require 10 or more, making it challenging to decide how many trials to allocate to each project. Second, these decisions carry high stakes. Each trial is both costly—estimated at around $1,800—and time-consuming, with an average duration exceeding 200 days, making exhaustive testing infeasible. Lastly, dynamic considerations are crucial: insights gained from early trials can sharpen forecasts of success rate for related projects, thus guiding subsequent allocation decisions.

We also note that the participating labs in PSI operated collaboratively rather than competitively. These labs were funded through NIH's U01 mechanism, a collaborative grant structure, minimizing competition for funding. Additionally, each major lab focused on distinct target pools defined by specific organisms or protein families, further reducing competition.

Information. Participating labs regularly shared the trial allocation decisions and outcomes with each other on a weekly or quarterly basis. A centralized information system was established and maintained to facilitate this information-sharing process. Consequently, when making decisions, each lab's information set included historical project and trial outcomes from both their own trials and those of other labs.

3.2. Data Description

Our primary dataset contains detailed information on a total of 707,179 trials on 325,343 projects conducted by 40 participating labs during the entire PSI, including trial start and end date, and trial outcome (failure or success).[4] Our empirical analysis focuses on dynamic decision making by the three largest labs—MCSG, NESG, and JCSG—during the production phase PSI-2. These three largest labs represent a major portion of our dataset, conducting 81.8% of total trials during PSI-2.[5]

Table 1 presents summary statistics of the primary dataset for projects and trials attempted by the three large labs in PSI-2. Note that, as mentioned above, these projects are highly risky, with project-level success rates below 4% for all three labs. The variation in trial success rates likely suggests each lab's focus on different types of projects: for example, projects attempted by JCSG succeeded, on average, after 83.67 failed trials, whereas MCSG projects succeeded after only 1.159 failures. This motivates our decision to estimate each lab's decision-making separately, using lab-specific choice sets and structural parameters to accommodate potential differences in their project focus and experimental approach. We also report summary statistics for PSI-1 in Appendix A.2, and we do not analyze decision making during this phase but use PSI-1 data to initialize the prediction model.

In addition, we construct a secondary dataset of project characteristics relevant to scientists' decision making. For each project, we collect 705 physicochemical properties used by scientists in evaluating projects and have been shown to predict project success, such as taxonomic classifications (UniProt Consortium 2021), amino acid composition metrics (Babnigg and Joachimiak 2010, Jahandideh et al. 2014), and instability measurements (Slabinski et al. 2007, Jaroszewski et al. 2008). Furthermore, to capture additional latent structural dependencies that may influence human judgment, we supplement these properties with high-dimensional project embeddings obtained from a pretrained Transformer model (ProtTrans; Elnaggar et al. 2022). These embeddings effectively encode 1,024 biophysical characteristics of protein sequences.

Lastly, we obtain binary indicators of each project's novelty and biomedical importance, along with measures for the extent to which a sequence is associated with specific organism

[4] From the raw data (Berman et al. 2017), we exclude a small subset of projects (less than 4%) due to reasons such as missing protein sequences needed for characterizing project features, likely duplication with extremely similar projects, or because they were attempted across multiple PSI phases. We also excluded 13.8% of the recorded trial entries—treating these as unattempted—because they never reached the initial wet-lab step (*cloned*), and thus had no verified lab effort. See Appendix A.1 for additional details.

[5] The remaining labs operated at a significantly smaller scale, and including them would introduce complexity of presentation without substantively adding new insights. One exception is NYSGRC, another large-scale center, which we exclude from analysis due to data availability constraints. (Specifically, for NYSGRC, over half of successful projects have missing or incorrectly ordered dates for key intermediate trial stages, preventing accurate modeling of scientists' dynamic decision-making processes.)

Table 1 Summary Statistics of the Primary Dataset in PSI-2

Definition	JCSG	MCSG	NESG	Others	Total
Project-Level Statistics					
Projects attempted	26,932	37,501	14,319	33,564	112,316
Projects succeeded	865	623	564	3,896	5,948
Project success rate (%)	3.212	1.661	3.939	11.61	5.296
Average number of trials per project	7.996	1.195	3.611	2.066	3.394
Average number of trials to failed projects	5.385	1.194	3.464	1.942	2.723
Average number of trials to succeeded projects	86.67	1.252	7.218	3.008	15.39
Average number of failed trials before success	83.67	1.159	6.748	1.563	13.95
Trial-Level Statistics					
Trials allocated	215,338	44,800	51,713	69,332	381,183
Trials to successful projects	74,968	780	4,071	11,718	91,537
Trials succeeded	929	680	620	8,043	10,272
Fraction of trials to successful projects (%)	34.81	1.741	7.872	16.90	24.01
Trial success rate (%)	0.431	1.518	1.199	11.60	2.695
Average trial duration in days	226.9	133.3	453.7	87.32	221.3

groups—namely, human proteins, eukaryotic proteins, and membrane proteins—as defined in NIH's evaluation reports (NIGMS 2012). We later utilize these metrics to broaden our discussion of the labs' objectives. See Appendix A.3 for details of the secondary dataset on project features, including the evaluation metrics. These project features facilitate cross-project learning in our models.

4. Machine vs. Human Decisions

In this section, we first formalize the dynamic innovation decision-making problem based on our empirical context. We then develop a machine learning algorithm based on the CN-UCB framework to solve this problem. Finally, we estimate a structural model of human decision making to understand scientists' trial-allocation behavior observed in the data.

4.1. Problem Setup

We focus our analysis on lab decision making during the production phase (PSI-2). This choice offers several advantages. First, it allows us to leverage prior trial information from the preceding PSI-1 phase, which provides essential initial information to avoid the cold-start problem. Second, PSI-2 had a publicly stated clear primary objective: determining as many unique protein structures (unique successful projects) as possible given labs' budget constraint. This clarity enables us to set up a clean and well-defined decision-making problem. (In contrast, PSI-3 shifted focus toward projects' biomedical importance, making evaluation complicated due to labs' unknown objective weights.) We set each period to one month, resulting in 60 periods during PSI-2.

Now, we formally define the objective function that the labs try to optimize. Suppose we have T periods and J projects in total. Over the study horizon ($t \in \{0, ..., T\}$), each lab i aimed to decide the

number of trials allocated to each project ($j \in \{1, ..., J\}$) in each period, $\boldsymbol{q_i} := \{q_{ijt}\}_{j=1,...,J,t=0,...,T}$, to maximize the expected number of unique successful projects:

$$\begin{aligned} &\max_{\boldsymbol{q_i}} \sum_t \sum_j \mathbb{E}_{\mathcal{F}_t}[\mathbb{S}\{q_{ijt}, \boldsymbol{X}_{ij}\}], \\ &\text{s.t. } \sum_j q_{ijt} \leq Q_{it}, \forall t \in \{1, ..., T\}, \end{aligned} \tag{1}$$

where $\boldsymbol{X}_{ij}$ is the vector summarizing the 1,729 features characterizing project j. $\mathbb{S}$ is a binary indicator for whether project j is successful (1) or not (0) under q_{ijt} number of trials allocated. Finally, $\mathcal{F}_t$ denotes the information set available at the start of period t, including the accumulated history of allocations $\{q_\tau\}_{\tau<t}$ and their corresponding outcomes among all participating labs in periods up to $t-1$. Thus, $\mathcal{F}_t$ informs lab beliefs about project success probabilities. As mentioned earlier, we focus our analysis on PSI-2, which includes 60 monthly periods. The initial information set, $\mathcal{F}_{t=0}$ includes all available data from PSI-1 and is used to initialize the model. The labs face resource constraints: in any period t, the total number of trials allocated to all the projects, $\sum_j q_{ijt}$, should not exceed the trial capacity Q_{it} of lab i in that period.

This formulation captures the labs' dynamic challenge: they balance exploitation—focusing on projects whose feature profiles already suggest a high probability of success—with exploration—investing in less-familiar projects to better learn the relationship between project features and success potential. We thus frame this decision-making process as a combinatorial contextual bandit: the project feature vectors provide the context, the allocation vector forms the combinatorial action, and the resulting trial outcomes generate rewards.

4.2. A Benchmark ML Algorithm

To benchmark human decision making, we implement a state-of-the-art ML policy grounded in the CN-UCB framework. Because each project is described by a rich set of biological and physicochemical features (more than 1,700), the commonly-used linear UCB methods (LinUCB; Li et al. 2010) are likely too restrictive to flexibly capture the underlying feature–success relationships. Thus, we build on recent advances that extend UCB algorithms to high-dimensional contexts (e.g., Hwang et al. 2023) and incorporate techniques from research on multi-armed bandits for resource allocation (e.g., Zuo and Joe-Wong 2021, Ge et al. 2024). The CN-UCB framework leverages deep neural networks to capture complex non-linear feature–success relationships and supports cross-project learning through shared latent representations. We formulate the lab's dynamic problem as a repeated single-period optimization.

In period t, the CN-UCB algorithm determines the number of trials allocated to project j for lab i, $\boldsymbol{q}_{it} := \{q_{ijt}\}_{j=1,..,J}$, by solving:

$$\begin{aligned} &\max_{\boldsymbol{q}_{it}} \sum_{j} u^{ML}_{\mathcal{F}_t}(q_{ijt}, \boldsymbol{X}_{ij}), \\ &\text{s.t. } \sum_{j} q_{ijt} \leq Q_{it}, \end{aligned} \tag{2}$$

in which $u^{ML}_{\mathcal{F}_t}(q_{ijt}, \boldsymbol{X}_{ij})$ is the upper confidence bound of the expected probability that at least one of the active trials on project j will succeed:

$$u^{ML}_{\mathcal{F}_t}(q_{ijt}, \boldsymbol{X}_{ij}) = \underbrace{1 - (1 - p^{ML}_{i,\mathcal{F}_t}(\boldsymbol{X}_{ij}))^{\tilde{q}_{ijt}+q_{ijt}}}_{\text{Immediate success}} + \underbrace{\gamma^{ML}_i B(\boldsymbol{\theta}^{ML}_{it}, \boldsymbol{X}_{ij}, q_{ijt})}_{\text{Exploration value}}. \tag{3}$$

Immediate success. The first term captures the predicted probability that at least one active trial on project j will succeed. Here, $p^{ML}_{i,\mathcal{F}_t}(\boldsymbol{X}_{ij})$ represents the predicted single-trial success probability for project j. This prediction is generated by a deep neural network, trained using the project's feature vector $\boldsymbol{X}_{ij}$ along with the accumulated historical trial-allocation and outcome data captured within the information set $\mathcal{F}_t$. We consider the trial outcomes as independent draws from a Bernoulli distribution parameterized by project characteristics.[6] A project is considered successful if at least one trial allocated to it succeeds. The sum $q_{ijt} + \tilde{q}_{ijt}$ denotes the total number of active trials on project j, where $\tilde{q}_{ijt}$ is the number of ongoing trials allocated to project j before period t, and q_{ijt} is the newly allocated trials in period t. This cumulative trial effort is considered because the outcomes of trials previously allocated to the same project are likely still pending when labs decide whether to allocate additional trials in each period (trials typically took over 200 days to complete, whereas each period in our model is one month).

Exploration bonus. The second term in Equation (3) encourages trials on projects where the network is more uncertain about the outcome. The exploration bonus $B(\boldsymbol{\theta}^{ML}_{it}, \boldsymbol{X}_{ij}, q_{ijt})$ is adapted from Hwang et al. (2023) and Faury et al. (2020) to account for multiple independent trials per project. It depends on $g(\boldsymbol{X}_{ij}, \boldsymbol{\theta}^{ML}_{it})$, the gradient of the network's output with respect to its parameters, and Z_{it}, the cumulative outer product of these gradients across all trials whose outcomes have been observed (see Appendix B for the full derivation). Intuitively, $B(\cdot)$ is large when the network is uncertain about a project—the gradient of the network's output with respect to its parameters signals how sensitive the model is to that project's inclusion, which increases the accumulated

[6] Labs typically allocated multiple trials to the same project simultaneously to explore a wide range of experimental conditions, such as the type and concentration of precipitants, pH, temperature (Chandonia et al. 2004, Chayen and Saridakis 2008). This conditional independence is also supported by the empirical data: for the three largest labs in our sample, 95.2% of projects received all trial allocations in a single, simultaneous batch during PSI-2. As these trials were initiated in parallel within the same period t, there was little opportunity for within-batch sequential updating based on realized outcomes.

information matrix and down-weights the exploration bonus. The term thus encourages trials on unfamiliar, high-uncertainty projects. A tuning parameter γ_i^{ML} governs the overall strength of exploration, and its optimal value is selected via grid search.

As mentioned above, we constructed a deep neural network to obtain $p_{i,\mathcal{F}_t}^{ML}(\boldsymbol{X}_{ij})$ and $g(\boldsymbol{X}_{ij}, \boldsymbol{\theta}_{it}^{ML})$. Specifically, it is a multi-layer-perceptron (MLP) with 3 hidden layers (widths: 512, 512, 8). In each period t, we train the network on all available historical trial outcomes and corresponding project characteristics, treating each trial as a binary classification task (success = 1, failure = 0). See Appendix C.1 for detailed model architecture and training process. We also show that our model's predicted success probabilities align well with the empirical project outcomes in Appendix C.2.

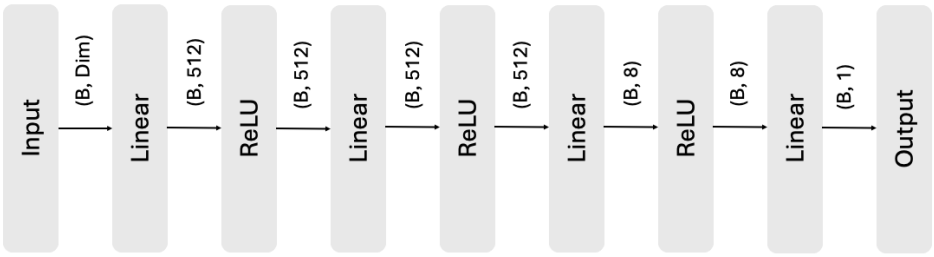


Figure 2 Neural Network Structure

Note: Figure 2 presents the structure of our neural network. It has three hidden layers, with a batch size of 4,096. The input dimension is 1,729. The hidden layers have a width of 512, 512, and 8 respectively, each followed by a ReLU activation layer.

4.3. A Structural Model of Human Decisions

We next propose a structural model to capture how scientists allocate trials in our data. We formulate their empirical behavior as a generalized, flexible variant of our CN-UCB benchmark policy. This structural framework is highly accommodating: whether scientists were rigorously optimizing based on ML predictions or relying on simpler cognitive heuristics to process complex project features, our model can capture their underlying policy by estimating how they internally weighted information and valued exploration.

Estimating this structural model enables us to perform the dynamic-contraction counterfactual analysis later, in which human allocation decisions must be inferred under a reduced trial capacity.

4.3.1. Utility Specification. To structurally characterize human behavior, we model the labs' utility function as a generalized version of CN-UCB benchmark algorithm.[7] This choice is

[7] The UCB framework serves as a natural foundation because it explicitly separates immediate expected rewards (beliefs) from the value of learning (exploration).

supported by recent empirical evidence demonstrating that scientists' dynamic allocation decisions in the PSI context closely track a UCB-style heuristic Zhuo (2025). In period t, lab i evaluates candidate project j with expected utility:

$$u^H_{\mathcal{F}_t}(q_{ijt}, \boldsymbol{X}_{ij}) = \underbrace{1-(1-p^H_{i,\mathcal{F}_t}(\boldsymbol{X}_{ij}))^{\tilde{q}_{ijt}+q_{ijt}}}_{\text{Immediate success}} + \underbrace{\gamma^H_i \beta^{H\,\tilde{q}_{ijt}+q_{ijt}-1}_i B(\boldsymbol{\theta}^H_{it}, \boldsymbol{X}_{ij}, q_{ijt})}_{\text{Exploration value}}, \tag{4}$$

where $p^H_{i,\mathcal{F}_t}(\boldsymbol{X}_{ij})$ represents the scientists' perceived single-trial success probability, modeled by a separate neural network parameterized by $\boldsymbol{\theta}^H$. The parameter γ^H_i governs the overall intensity of exploration by lab i. Lastly, $\beta^H_i \in [0,1]$ is a decay term, controlling the marginal exploration value of additional trials on the same project. We introduce the decay term because some of our labs adopted a "high-throughput" approach, in which they would initially assign one trial to a large number of projects, and then decide whether to add more trials based on the results of these earlier trials (NIGMS 2009). As a result, humans may exhibit different valuations on the exploration value of single trials compared to our algorithm. At the extremes, $\beta^H_i = 1$ implies the same discounting as in the ML benchmark, whereas $\beta^H_i = 0$ implies only the first trial on each project carries exploration value.

Comparing Equation (3) and Equation (4), the human policy deviates from CN-UCB along three key dimensions: (1) success-rate predictions ($p^{ML}_{i,\mathcal{F}_t}$ vs. $p^H_{i,\mathcal{F}_t}$), (2) overall exploration intensity (γ^{ML}_i vs. γ^H_i), (3) within-project exploration decay (1 vs. β^H_i).

4.3.2. Transfer-Learning to Reduce Parameter Space of $\boldsymbol{\theta}^H$. In CN-UCB, all neural-network parameters are estimated via supervised learning; by contrast, the human parameters $\boldsymbol{\theta}^H$ need to be identified through structural estimation, which relies on economically interpretable variation in human actions across features. Hence, full identification of the 1.1 million parameters in $\boldsymbol{\theta}^H$ is infeasible.

We address this with a transfer-learning strategy that is both principled and practical. The key idea is that the hidden layers of the neural network capture a general, low-dimensional representation of project characteristics—a learned feature extraction that should be largely invariant across human and algorithmic processing of the same data. What differs between humans and the algorithm is how they *weight* these representations when forming beliefs about success probabilities.[8] We therefore freeze all hidden-layer parameters at the CN-UCB values and allow the human network to differ only in the output layer (in particular, 9 parameters: 8 weights and a bias term) from the ML neural network. We denote these free weights by $\boldsymbol{\theta}^{Ho}_{it}$.

[8] While we do not observe the exact ML model the scientists used for any specific decision, our transfer-learning approach accommodates these possibilities: by estimating flexible weights on a shared, high-dimensional latent feature space, our structural model can approximate this wide range of unobserved human predictive strategies.

Specifically, the estimation proceeds in two steps (see Figure 3). In Step 1, we fit an auxiliary neural network—with the same architecture as the CN-UCB model—on empirical human trial outcomes, obtaining parameters $\boldsymbol{\theta}_{it}^{A}$. This yields, for each project j, a low-dimensional representation $\boldsymbol{S}_{ijt}^{A} \in \mathbb{R}^{1\times 9}$ as the input to the output layer. In Step 2, we replace the auxiliary network's output-layer weights $\boldsymbol{\theta}_{it}^{Ao}$ with the human-specific weights $\boldsymbol{\theta}_{it}^{Ho}$, obtained through $p_{i,\mathcal{F}_t}^{A}(\boldsymbol{X}_{ij}) = \sigma(\boldsymbol{S}_{ijt}^{A}\boldsymbol{\theta}_{it}^{Ao})$, in which $\sigma(\cdot)$ is the sigmoid transformation. Intuitively, we assume that provided with the same training data, humans and the auxiliary algorithm observe the same low-dimensional representations of the characteristics for each project (i.e., $\boldsymbol{S}_{ijt}^{A} = \boldsymbol{S}_{ijt}^{H}$), but they may assign different weights to them (i.e., $\boldsymbol{\theta}_{it}^{Ao}$ and $\boldsymbol{\theta}_{it}^{Ho}$) when forming beliefs on the success rate.

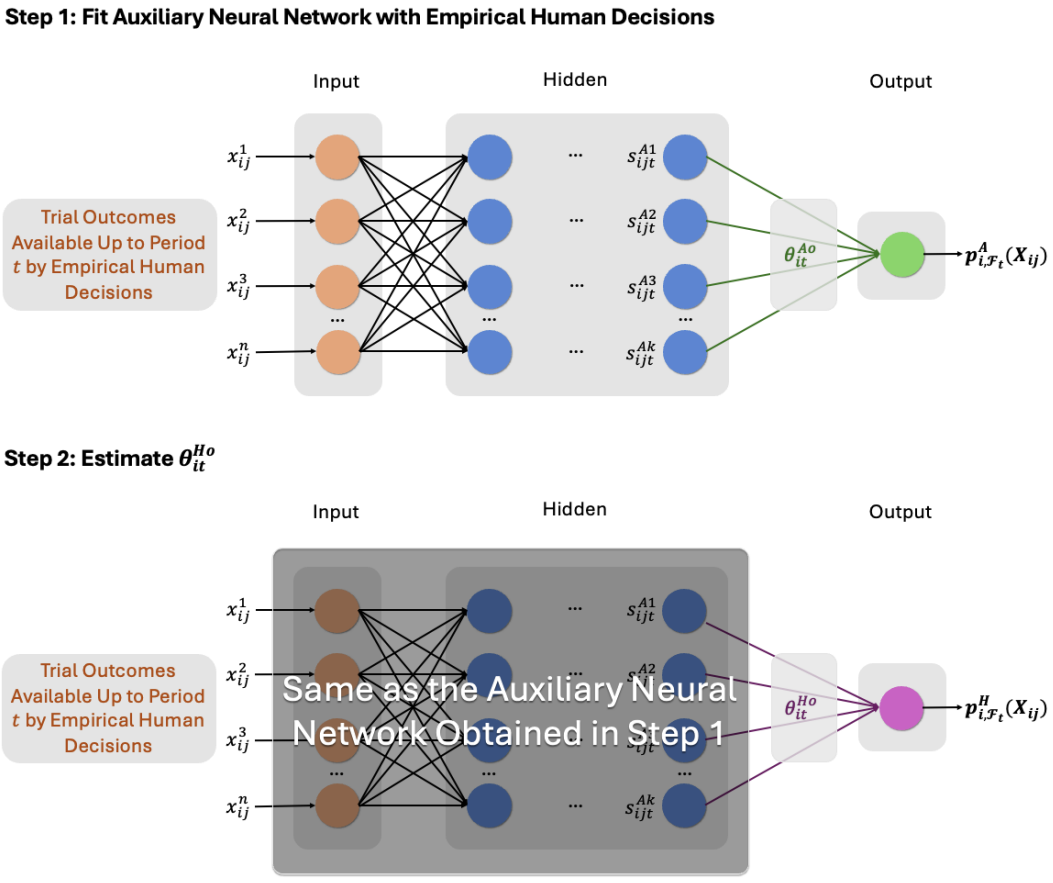


Figure 3 Procedure to Estimate the Parameters in the "Human Network"

Note: Figure 3 presents the estimation procedure for $\boldsymbol{\theta}_{it}^{Ho}$, the output layer parameters of the "Human Network". We first fit an auxiliary neural network with the data from empirical human decisions. Then, in Step 2, we freeze the input and hidden layers of the auxiliary neural network, and estimate $\boldsymbol{\theta}_{it}^{Ho}$ by calculating $p_{i,\mathcal{F}_t}^{H}(\boldsymbol{X}_{ij})$.

4.3.3. Identification and Estimation. The three sets of parameters $\boldsymbol{\theta}_{it}^{Ho}$, γ_i^H and β_i^H are identified from distinct, economically interpretable sources of variation in the empirical trial allocation data. Below we briefly discuss the intuition. $\boldsymbol{\theta}_{it}^{Ho}$ is identified by variations in allocations among

projects that have similar exploration values but differ in their feature vectors. Intuitively, when two projects offer the same exploration bonus, differences in allocation would reflect differences in predicted success rates—which are governed by $\boldsymbol{\theta}_{it}^{Ho}$. γ_i^H is identified by variations in the decision to attempt or skip projects that have similar predicted single-trial success rates but differ in their exploration bonuses. When two projects have the same predicted payoff but one is more uncertain, the relative willingness to attempt the uncertain project reveals the weight placed on exploration. β_i^H is identified by variations in the number of trials allocated to projects once selected, specifically when projects have similar initial predicted success probabilities. The rate at which scientists add trials to the same project (versus switching to new projects) reveals how quickly they discount the informational value of additional trials.

Leveraging the empirical human trial allocation decisions, we can use the likelihood for the observed human behaviors to estimate $\boldsymbol{\theta}_{it}^{Ho}$, γ_i^H, and β_i^H. Specifically, we denote $W_{ijt}^k = 1$ if trial k to project j was allocate by lab i in period t, and $W_{ijt}^k = 0$ otherwise. Then, for each trial that could be allocated in period t, we can obtain $V(W_{ijt}^k = 1|\boldsymbol{\theta}_{it}^{Ho}, \gamma_i^H, \beta_i^H)$. The incremental utility of launching this trial is modeled as:

$$V(W_{ijt}^k = 1|\boldsymbol{\theta}_{it}^{Ho}, \gamma_i^H, \beta_i^H) = u_{\mathcal{F}_t}^H(q_{ijt} = k, \boldsymbol{X}_{ij}) - u_{\mathcal{F}_t}^H(q_{ijt} = k-1, \boldsymbol{X}_{ij}) + \epsilon_{ijkt}, \tag{5}$$

where ϵ_{ijkt} follows a Type-I extreme-value distribution, allowing us to obtain a standard logit probability of lab i allocating the k-th trial to project j in period t, denoted as $P(W_{ijt}^k = 1|\boldsymbol{\theta}_{it}^{Ho}, \gamma_i^H, \beta_i^H)$ (Note that to keep the model parsimonious and estimation feasible, we do not enumerate choice probabilities for every bundle of project–trials.)[9] The parameters are estimated by maximizing the log-likelihood of the observed trial-allocation decisions over all candidate trials and periods:

$$\hat{\boldsymbol{\theta}}_{it}^{Ho}, \hat{\gamma}_i^H, \hat{\beta}_i^H := \arg\max \sum_t^T \sum_j^J \sum_k [\underbrace{W_{ijt}^k \log\big(P(W_{ijt}^k = 1)\big)}_{\text{Empirically allocated trials}} + \underbrace{(1 - W_{ijt}^k) \log\big(1 - P(W_{ijt}^k = 1)\big)}_{\text{Empirically not allocated trials}}]. \tag{6}$$

We then estimate $\boldsymbol{\theta}_{it}^{Ho}$, γ_i^H and β_i by maximizing the above likelihood function. See Appendix D for details on how we construct $P(W_{ijt}^k = 1|\boldsymbol{\theta}_{it}^{Ho}, \gamma_i^H, \beta_i^H)$ and the estimation procedure.

4.3.4. Estimation Results. We estimate the above structural model using human empirical decision making data during PSI-2. In period $t = 0$, we use project characteristics and history trial outcome information available at the start of PSI-2 to initialize our auxiliary neural network. Table 2 presents the estimated human exploration parameters. Columns (1)-(3) show $\hat{\gamma}_i^H$, the estimated parameter governing exploration intensity, and Columns (4)-(6) present the $\hat{\beta}_i^H$, the estimated

[9] Specifically, we form probabilities at the project-trial level: a trial is selected when its incremental utility exceeds an "outside-option" threshold—the utility of the last trial that fits within that period's capacity. See Appendix D for estimation details.

parameter governing how humans discount the exploration values of single trials allocated to the same project. Note that for all three labs, $\hat{\beta}_i^H$ is smaller than 1, indicating that compared to the algorithm, humans tend to discount the exploration value of an additional trial allocated to the same project more heavily. Due to space constraints, detailed estimation results for the "Human Network" parameters $\boldsymbol{\theta}^{Ho}$ for each lab over time are reported in Appendix D.

Table 2 Estimation Results of Human Exploration Parameters

Exploration Intensity ($\hat{\gamma}_i^H$)			Exploration Decay ($\hat{\beta}_i^H$)		
JCSG	MCSG	NESG	JCSG	MCSG	NESG
(1)	(2)	(3)	(4)	(5)	(6)
0.0005086	0.004036	0.01295	0.9899	0.8576	0.9553
(7.28e-08)	(9.93e-07)	(5.79e-06)	(1.80e-06)	(1.44e-05)	(6.32e-06)

Note: Standard deviations in parentheses.

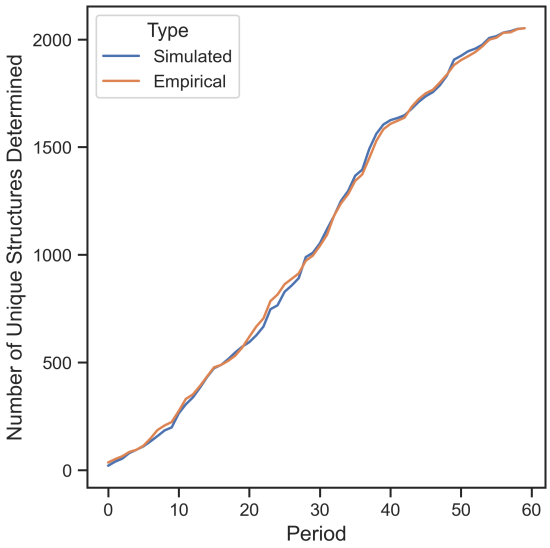


Figure 4 Simulated and Empirical Allocation Results

Note: Figure 4 presents the cumulative number of unique structures determined over time from the empirical and simulated allocations. During the simulation process, due to the *selective-label bias* (Lakkaraju et al. 2017), only project–trial pairs with empirically observed results can be allocated. The choice set in each period is limited to projects that humans had already attempted or would attempt within the next six periods. Trial outcomes and durations replicate the exact empirical sequence.

4.3.5. Model Fit. To assess model fit, we simulate human allocations over the entire decision horizon in PSI-2 using the estimated human parameters under the same empirical trial capacity. Figure 4 plots the cumulative trajectory of unique structures obtained over the 60 periods: the blue line shows outcomes from the simulated model, and the orange line shows the true empirical

trajectory. The two curves track closely, indicating that the estimated structural model accurately replicates human allocation behavior as well as its corresponding outcomes. See Appendix E for further details on the simulation procedure, along with additional model-fit evaluations (including trajectories for total projects attempted and distributions of trial allocations across projects).

5. Machine vs. Human Comparison

In this section, we compare the performance of our benchmark algorithm against human decision makers. Both policies are initialized with PSI-1 information (including project characteristics and historical trial outcomes), and their dynamic decision-making performance is evaluated over PSI-2.

A direct comparison is complicated by *selective-label bias*: trial outcomes are unknown for project–trial pairs that humans never attempted, so allocating additional trials beyond the empirical record in simulation could yield misleading counterfactuals (Lakkaraju et al. 2017). This is because scientists may have relied on information not captured in our data; any simulated performance gap might therefore reflect missing data rather than true decision-making inefficiency. We address this challenge in two steps. First, we present a quasi-contraction analysis that reveals a performance gap under a common trial budget but cannot fully rule out alternative over-time preference explanations (Section 5.1). Second, we discuss our dynamic contraction method—the paper's central methodological contribution—which provides a credible dynamic comparison (Section 5.2).

To ensure a fair comparison under both approaches, we restrict all policies to a *common observed outcome space*: only project–trial pairs with empirically observed results are eligible for allocation. Operationally, we add a constraint on the per-period trial-allocation problem (Equation (2)):

$$\sum_{t} q_{ijt} \leq Q_{ij}, \forall j \in \{1, ..., J\}, \tag{7}$$

in which Q_{ij} is the total number of trials allocated to project j empirically by lab i in the data. This cap ensures that any simulated policy never allocates more trials to a project than were observed empirically. By leveraging the one-sided nature of the data—that is, outcomes are known only for actions actually taken—all policies under evaluation operate exclusively within the observed outcome space. Thus, while trial actions are simulated according to each policy, the corresponding trial outcomes are read directly from the empirical record, with no reliance on simulated or unobserved results.

Simulation details. We simulate the decision and outcome trajectories for both human and ML policies as follows. At the start of each period t, we update the training data for each policy with newly revealed trial outcomes and retrain the corresponding neural network model (the deep neural network from Figure 2 for the ML policy and the human network from Figure 3 for the human policy). Each policy then allocates trials in period t by solving Equation (2), where the human

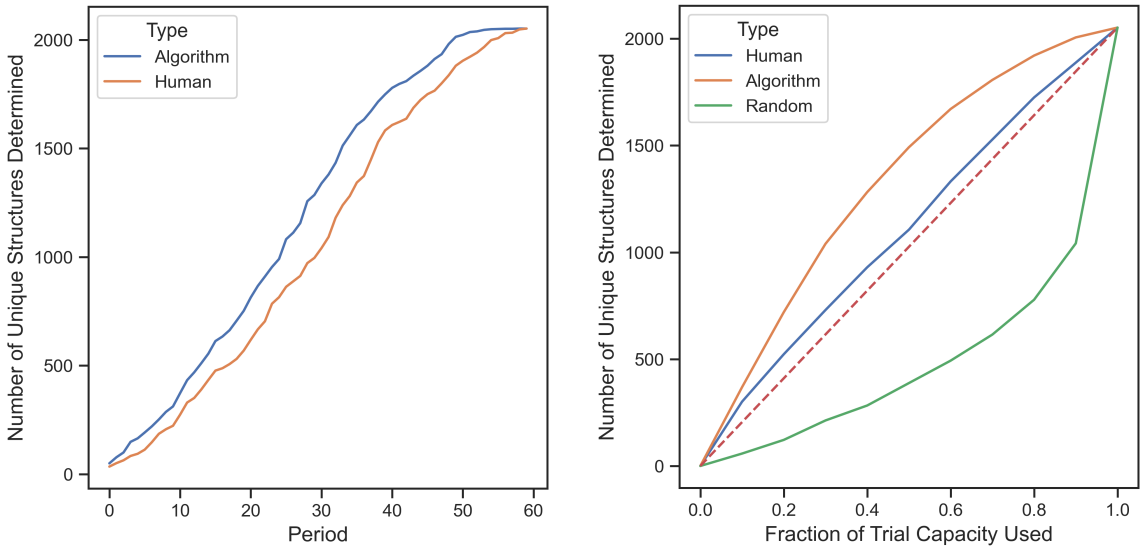
policy replaces u^{ML} with its counterpart u^{H}. To reflect the evolving project assignment pool, the choice set in t is limited to projects that had been attempted empirically up to and including period t, or would be attempted within the next six periods (results are robust to this window). Trial outcomes and durations are read directly from the empirical sequence. For example, if a project historically succeeded only on its tenth trial, the simulation records success only after nine failures. If a trial outcome was empirically revealed after six periods, it is incorporated into the neural network training data during simulation with the same outcome revelation duration (i.e., six periods after its trial allocation). These realized outcomes from each policy's allocations then update the information set for their respective neural networks at the start of the next period $t+1$. See Appendix E.1 for more details on the simulation procedure.

5.1. Quasi-Contraction: Revealing the Performance Gap

As a first step, we ask: If the model had exactly the same period-by-period trial budget as the human labs, how would it rearrange trials differently over time? Human performance comes directly from the empirical record. To simulate the benchmark algorithm, we let it allocate the same number of trials per period as the labs empirically did, $t = 0$ through T, during PSI-2. Because all trials are drawn from the common observed outcome space, the algorithm essentially reshuffles them—moving some allocations earlier or later than the humans did.

Figure 5(a) compares cumulative unique structures over time achieved by the benchmark algorithm (blue) and the empirical human policy (orange). The two curves converge at the final period T by construction, because the algorithm has exhausted exactly the same set of trials that the labs empirically allocated during PSI-2, yielding identical cumulative performance at period T. In every intermediate period t $(0 < t < T)$, the algorithm consistently outperforms humans: drawing a vertical line anywhere between the first and last periods reveals that the algorithm achieves more unique structures cumulatively compared to human labs. For example, at period 30, the human policy obtains 1,041 unique structures while the algorithm obtains 1,339—a 28.6% increase. This result implies that if the resource allocation process were halted at any period between 1 and $T-1$, the benchmark algorithm would deliver higher cumulative success.

This analysis reveals a substantial performance gap, but it has an important limitation. Because both policies converge at the final period T by construction (they exhaust the same total trial set), the quasi-contraction cannot rule out the possibility that scientists were intentionally patient—forgoing early successes in favor of exploration that pays off at the horizon. Moreover, the result establishes that the algorithm dominates at every interim point, but it cannot demonstrate dominance at the terminal point. Addressing this requires a different comparison strategy.



(a) Results of Quasi-Contraction

(b) Results of Dynamic Contraction

Figure 5 Results of Machine vs. Human Comparison

Note: Figure 5(a) presents the results of Quasi-Contraction. The x-axis is period, and the y-axis is the cumulative number of unique success obtained by our algorithm and empirical human allocation. In each period, the algorithm allocates the same number of trials as humans empirically did during PSI-2. In every intermediate period, the algorithm consistently obtains more unique successful structures than humans. Figure 5(b) presents the results of Dynamic Contraction. In this comparison, we progressively reduce the trial capacity by a retention fraction $x \in \{100\%, 90\%, 80\%, ..., 0\%\}$ across all periods. Then, we compute outcomes under three trial-allocation policies at each capacity level: (1) the benchmark CN-UCB algorithm, (2) the human structural model, and (3) a random baseline. Unlike Figure 5(a), this plot does not track time along x-axis—each point reflects the end-of-PSI-2 overall result at that capacity level. The y-axis is the number of unique success obtained.

5.2. Dynamic Contraction: Overcoming Selective Labels in a Dynamic Setting

To provide a credible comparison that is not confounded by horizon effects, we introduce the dynamic contraction method. The key idea is to progressively reduce the total trial capacity available to both policies and compare their terminal performance at each capacity level. If the algorithm achieves more unique structures than the human policy at the end of the full horizon under a tighter budget, then the human policy is definitively outperformed—patience cannot explain the gap, because both policies have the same time horizon and the same (reduced) resources.

Implementing this comparison for the benchmark ML algorithm side is straightforward: we solve the CN-UCB optimization with a scaled-down capacity constraint. The challenge is on the human side: we must infer how scientists would have allocated trials under a budget they never actually faced. This is where the structural model becomes essential. Using the estimated human parameters, we simulate how scientists would reallocate trials when their per-period capacity is reduced to a fraction x of the original—capturing scientists' sequential actions and evolving information accumulation, and exploration behavior under the counterfactual constraint.

Specifically, we progressively reduce the trial capacity across all periods for both humans and the algorithm. At each capacity level (e.g., 70% of the original capacity), the human structural model and the benchmark algorithm select trial allocations from the common observed outcome space, following their respective policies described earlier. This procedure generates complete decision trajectories for both policies under the same resource constraints, while capturing their distinct information accumulation processes driven by their respective period-by-period trial allocation decisions. Note that at full (100%) capacity human and algorithmic allocations coincide by construction (both become the actual empirical allocations); meaningful differences emerge only when capacities are progressively contracted.

We now present the dynamic-contraction results. We progressively reduce the trial capacity by a retention fraction $x \in \{100\%, 90\%, 80\%, ..., 0\%\}$ across all periods. As x decreases, the capacity tightens proportionally, forcing both policies to prune their allocations from the common observed outcome pool. At each capacity level, we compute outcomes under three distinct trial-allocation policies: (1) the benchmark CN-UCB algorithm, (2) the human structural model, and (3) a random baseline, which randomly retains a fraction x of the empirically observed human-allocated trials in each period. Figure 5(b) plots the results: each point reflects end-of-PSI-2 unique structures obtained at a given capacity level (unlike Figure Figure 5(a), the x-axis is capacity fraction, rather than time). For example, if we draw a vertical line at $x = 0.7$, its intersections with the three solid lines shows how many unique structures each policy achieves throughout PSI-2 when limited to 70% of the full trial capacity in every period.

Below summarizes the key observations. Most importantly, the benchmark ML algorithm consistently outperforms the human decisions at all trial-capacity levels, reflecting the algorithm's superior ability to identify promising projects and efficiently allocate trials. For example, at 70% of original trial capacity, the algorithm achieves significantly more (18.3%) unique protein structures than the human policy. We note two additional observations. First, the random baseline consistently underperforms, falling well below the diagonal line (representing proportional success). This underscores that effective trial allocation requires carefully choosing appropriate trial "depth," which random selection fails to achieve.[10] Second, although the human structural model performs worse than the ML benchmark, it significantly outperforms random selection.

5.3. Overall Difference

We confirm that the benchmark algorithm yields significantly more unique successful projects compared to all three labs. Specifically, Panel A of Table 3 presents the number of unique structures

[10] For intuition, imagine every project needs ten trials to succeed. A random policy that spreads trials thinly across many projects may never give any single project its tenth trial, so very few successes are realized.

obtained by humans and the benchmark algorithm. Overall, the algorithm obtains 18.3% more unique structures than the human scientists. Moreover, while humans and the algorithm allocate the same number of total trials, their distribution of trials per project differs markedly. Panel B of Table 3 compares the number of projects attempted by the machine and humans. The results show that scientists spread their efforts across a broader set of projects (59,576 in total), resulting in fewer trials per project; in contrast, the algorithm invests greater "depth" into fewer, more selective projects (50,392 in total).

Table 3 Comparison of Human and Algorithm Policies: Overall Performance and Project Attempted

Panel A	Unique Structures Determined			
	JCSG	MCSG	NESG	Total
Human Prediction + Human Exploration	533	568	427	1,528
Algorithm Prediction + Algorithm Exploration	682	603	522	1,807
Improvement (%)	28.0	6.16	22.2	18.3
Panel B	Number of Projects Attempted			
	JCSG	MCSG	NESG	Total
Human Prediction + Human Exploration	20,500	26,760	12,316	59,576
Algorithm Prediction + Algorithm Exploration	14,842	25,560	9,990	50,392
Choice Set	26,932	37,501	14,319	78,752

In our analysis, we assume that the labs' objective during PSI-2 is to maximize the total number of unique successful protein structures. Although this goal was publicly stated, labs may also have weighed other considerations—such as project novelty and biomedical importance—when allocating trials. If the benchmark algorithm obtains additional successful structures at the cost of these potential considerations, we cannot conclude that our algorithm improves human decisions. To alleviate this concern, we examine additional evaluation metrics, including the counts of unique successful structures across five categories: novel, biomedically important, human, membrane, and eukaryotic proteins (recall from Section 3). See Table 4 for the comparison results.

Table 4 Comparison Results with Expanded Objective Considerations

	Unique Structures Determined					
	Total	Novel	Biomedical	Human	Membrane	Eukaryotic
Human	1,528	952	814	33	110	51
Algorithm	1,807	1,183	938	44	121	61
Improvement (%)	18.3	24.3	15.2	33.3	10.0	19.6

As shown in Table 4, in addition to obtaining more unique structures, our algorithm also obtains more novel and biomedically important structures, as well as more structures related to certain

organisms compared to humans. Therefore, the benchmark algorithm indeed outperforms humans and can improve human decisions, even when the objective function is expanded and additional evaluation metrics are considered.

6. Why Does the Machine Outperform Human Scientists?

A natural question arises as we have established that the algorithm outperforms humans: Why does the machine systematically outperform human scientists in such dynamic decision-making? In this section, we discuss potential sources of human inefficiency. To illustrate, we focus on comparisons at a capacity retention fraction of $x = 70\%$ in dynamic contraction; results at other capacity levels are directionally similar.

6.1. Sources of Inefficiency

The previous subsection established that the benchmark algorithm delivers a robust performance gain over human decisions. We further demonstrate how to use it as benchmark to diagnose where human inefficiency arises, focusing on two distinct aspects: systematic prediction errors and sub-optimal exploration behavior. To clearly quantify each one, we examine each in turn: we match the human decision policy to the algorithm's along one aspect at a time—first correcting prediction accuracy, then exploration behavior—and measure the resulting improvements.

6.1.1. Systematic Prediction Errors. To isolate the inefficiency due to prediction error, we simulate human decision making with the estimated human exploration parameters (i.e., $\hat{\gamma}_i^H$ and $\hat{\beta}_i^H$), but use predictions on the success rate of projects generated by the ML algorithm. Table 5 compares the simulated outcomes with human and algorithm predictions. Specifically, if humans exhibit the same prediction of success rate of projects as the algorithm while maintaining their exploration levels, the total number of unique structures determined can increase by 13.2%, recovering about 72% ($= \frac{13.2\%}{18.3\%}$) of total improvements of our algorithm over humans. This result highlights that prediction error significantly contributes to human inefficiency, yet addressing it alone is insufficient to fully close the performance gap.

Table 5 Performance Comparison Under Human vs. Algorithmic Prediction

	Unique Structures Determined			
	JCSG	MCSG	NESG	Total
Human Prediction + Human Exploration	533	568	427	1,528
Algorithm Prediction + Human Exploration	647	588	495	1,730
Improvement (%)	21.4	3.52	15.9	13.2

6.1.1.1. Systematic Difference in Predictions. Beyond quantifying the aggregate performance loss from prediction errors, we directly compare the success-rate predictions produced by humans and by the algorithm to uncover any systematic biases in human predictions.

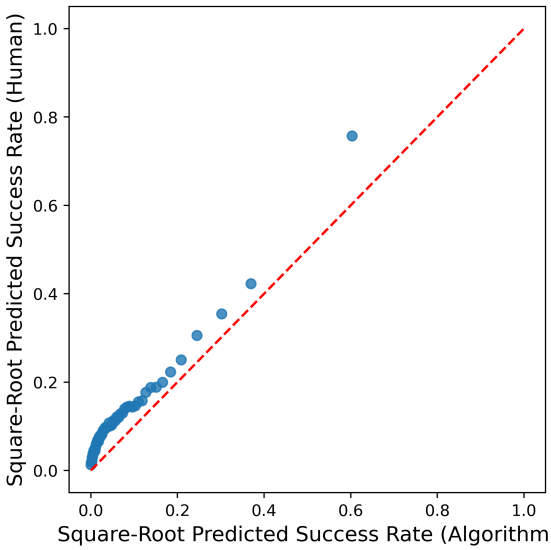


Figure 6 Human Predictions vs. Algorithm Predictions

Note: Figure 6 compares the human and algorithm predictions for projects attempted in PSI-2. We train the "Human Network" and our benchmark algorithm leveraging the information set at the end of PSI-2 obtained from empirical trial allocations. Then, we bin the projects into 50 quantile groups based on the algorithm predicted success rate, and compare against the mean predicted success rate by humans. Because the human and algorithm forecasts are on different scales, we compare them after min–max normalizing each lab's predictions to [0,1], preserving rank but removing scale effects.

We study their differences when both are provided with the same information set. To do so, we leverage the information set obtained through empirical trial allocations at the end of PSI-2 to train the "Human Network", and then train the benchmark algorithm using the identical data. We then compare the normalized human and algorithm predictions for the projects during PSI-2. (Note that we min–max normalize the algorithm predictions and those of each human lab to a [0,1] interval, before comparing them. This normalization ensures that observed differences reflect prediction biases rather than arbitrary prediction scale differences.)

Figure 6 shows the comparison between human and algorithm predictions. In particular, we bin the projects into 50 quantile groups based on the normalized algorithm predicted success rate, and compare with the normalized predicted success rate by humans. Because success rates are very low for most projects, we take the square root of the normalized predicted and actual success rates for better visualization. The results indicate that humans tend to overestimate the success rate of projects. This pattern aligns closely with established behavioral findings that people tend to

overweight low-probability events (e.g., Tversky and Kahneman 1979, Prelec 1998). We present the comparison of human predictions vs. algorithm predictions with the intermediate information sets during PSI-2 in Figure EC.6 in Appendix F, which shows largely consistent results.

6.1.2. Suboptimal Exploration Behavior. The above results show that correcting algorithm predictions recovers about 72% of total improvements of our algorithm over humans. Yet, it remains unclear if the scientists are exploring optimally with their own human predicted success rates. To answer this question, we simulate human decision making with the estimated "Human Network" parameters (i.e., $\hat{\boldsymbol{\theta}}_{it}^H$) so that humans retain their predictions on success rates, but use the optimal exploration parameters instead of the estimated $\hat{\gamma}_i^H$ and $\hat{\beta}_i^H$. Specifically, we set the optimal discounting factor, $\tilde{\beta}_i^H$ to equal to 1, and obtain the optimal exploration intensity $\tilde{\gamma}_i^H$ through grid search. Note that this exploration parameter may differ from the original CN-UCB one because it is re-optimized using human (rather than algorithm-generated) predictions. Then,

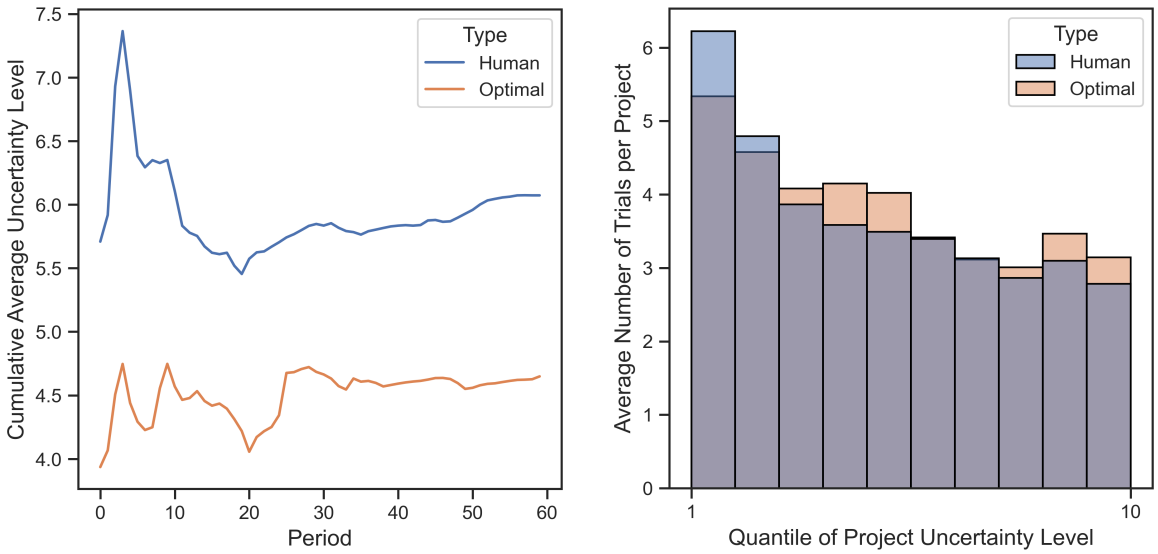
Table 6 Performance Comparison Under Human vs. Optimal Exploration Intensity

	Unique Structures Determined			
	JCSG	MCSG	NESG	Total
Human Prediction + Human Exploration	533	568	427	1,528
Human Prediction + Optimal Exploration	571	602	450	1,623
Improvement (%)	7.13	5.99	5.39	6.22

we simulate their trial allocation decisions with the optimal exploration levels and present the results in Table 6. The results indicate that with the optimal exploration levels, the total number of unique structures determined could increase by 6.22%, recovering approximately 34% ($= \frac{6.22\%}{18.3\%}$) of the algorithm's overall advantage. This result demonstrates that suboptimal exploration is indeed a significant contributor to human inefficiency, highlighting the complementarity between belief formation and exploitation-exploration behavior in driving overall dynamic performance.

6.1.2.1. Project-Selection Comparison. We compare exploration behavior by humans and by the algorithm while holding predictions fixed at the human ones. We first examine how the two policies differ in project selection across uncertainty levels. Recall that the magnitude of the term $g(\boldsymbol{X}_{ij}, \boldsymbol{\theta}_{it})$ in our algorithm signals overall how "uncertain" the neural network is about project j. Leveraging this measurement, we are able to quantify the uncertainty level of humans and the algorithm to each project. We bin the projects in the choice set into 10 quantile groups based on the uncertainty level derived from the information set in each period, and compare the cumulative average uncertainty level of projects attempted over time. Figure 7(a) presents the results. The x-axis indicates period, and the y-axis is the cumulative average uncertainty level of

projects attempted in each period (Q1 = most familiar (low-uncertainty), Q10 = most unfamiliar (high-uncertainty)). The results show that throughout PSI-2, scientists consistently over-explore in project selection: they select too few low-uncertainty projects and too many high-uncertainty ones compared to the optimal level.



(a) Uncertainty Levels of Projects Attempted Over Time

(b) Trial Depth Across Uncertainty Levels

Figure 7 Differences Between Human and Optimal Exploration Strategies

Note: Figure 7(a) presents the cumulative average uncertainty level of projects attempted over time by the human versus the optimal exploration strategies (both under human predictions). The uncertainty levels (Q1 = most familiar (low-uncertainty), Q10 = most unfamiliar (high-uncertainty)) are calculated based on the information set at the start of each period, leveraging $g(\boldsymbol{X}_{ij}, \boldsymbol{\theta}_{it})$ in our neural network. Figure 7(b) compares the average number of trials per project across uncertainty levels between the human and the optimal exploration strategies. The x-axis is uncertainty levels, which is calculated based on the information set at the start of PSI-2.

6.1.2.2. Trial-Depth Comparison. In addition to project selection, we also examine how improving human exploration affects the "depth" of trial allocation across projects at different uncertainty levels. Figure 7(b) presents the results. In particular, we bin all the projects in PSI-2 into 10 quantile groups based on the uncertainty levels at the start of PSI-2, and compare the number of projects with different uncertainty levels attempted by humans and our algorithm. The x-axis indicates the quantile of the uncertainty level of projects (Q1 = most familiar (low-uncertainty), Q10 = most unfamiliar (high-uncertainty)), and the y-axis is the average number of trials allocated to projects in this quantile. The results show that compared to the optimal exploration strategy, on average the scientists allocate more trials to relatively familiar projects, while they fail to "delve deep" into projects with higher uncertainty levels. One plausible explanation is that compared

to the algorithm, humans discount the exploration values of additional trials allocated to the same project more heavily (recall that $\beta_i^H < \beta_i^{ML} = 1$ for all three labs). As a result, when it comes to a project with high uncertainty, they are less likely to invest as many trials to it as our algorithm because the exploration values of additional trials diminish very quickly. Together with findings from Section 6.2.2.1, these results highlight that although scientists over-explore unfamiliar projects, they often spread effort too thinly rather than focusing deeply on the most promising opportunities.

7. Discussion and Conclusion

In high-stakes sequential environments, improving the quality of human decisions remains a core managerial challenge. This paper contributes a methodological framework for evaluating and diagnosing dynamic decision-making when only historical human-generated data are available. Specifically, we develop a dynamic contraction approach that combines structural modeling with algorithmic benchmarking to produce credible, one-sided comparisons between sequential decision policies under selective labels, delayed feedback, and path dependence. By decomposing performance gaps into distinct behavioral components—belief miscalibration and exploration inefficiency—the framework moves beyond aggregate human–algorithm comparison to provide actionable diagnostic insights for dynamic decisions.

We operationalize this framework in a rare, high stakes field setting—the $1.3 billion NIH Protein Structure Initiative (PSI)—by analyzing how scientists allocate costly experimental trials in a high dimensional, sequential problem. We find that, although PSI laboratories were led by experienced investigators and equipped with ample resources, their allocation choices were consistently outperformed by a CN-UCB policy that optimizes prediction and exploration simultaneously. Under identical budgets, the algorithm produced 18.7% more unique protein structures, and the advantage persisted when success was redefined to emphasize novelty, biomedical relevance, or specific structural classes. Using a structural model of scientists' decisions within our dynamic contraction framework and benchmarking against the ML policy, we identify two complementary sources of human inefficiency. Scientists systematically overestimated project success probabilities and over-explored by spreading trials thinly across too many new projects. Counterfactual simulations indicate that correcting belief miscalibration alone would recover about 72% of the algorithm's advantage, while improving exploration intensity alone would recover about 34%; closing the performance gap therefore requires improvements on both dimensions. More broadly, our results illustrate how machine learning can be used not merely as a substitute for human decision-makers, but as a diagnostic benchmark for identifying which components of human judgment should be improved in high-stakes sequential environments.

Our study speaks to a broad class of ML-augmented decision support systems in which organizational actors remain accountable for action selection but rely on algorithmic guidance. Many settings studied in the information systems literature—such as adaptive experimentation, dynamic pricing, platform governance, and R&D portfolio management—involve dynamic decision-making under the same combination of selective labels, delayed feedback, and sequential learning that motivates our framework. In such environments, particularly high-stakes ones where full automation is infeasible, the key challenge is often not only how to design a better predictive model, but also how to evaluate the incumbent human workflow and identify where algorithmic support would add the most value (Bryce McLaughlin 2024, Zhe Shen 2025). Our dynamic contraction approach enables practitioners to compare decision policies on a common observed outcome space and trace performance gaps back to specific behavioral margins, such as miscalibrated beliefs or inefficient exploration. This upstream diagnostic capability complements the design-oriented stream of human–AI research: before prescribing how humans and algorithms should collaborate, one must first understand which dimensions of human judgment need augmentation.

We conclude by highlighting several avenues for future research. First, our analysis primarily focuses on a single, production-phase objective and demonstrates robustness to alternative objectives. Explicitly investigating settings with multiple coexisting or evolving goals throughout the innovation process would offer valuable additional insights. Second, we examine scientists' decisions during PSI-2, when preliminary information was already available; understanding how laboratories initially select projects under severe uncertainty (the "cold-start" problem) and diagnosing decision inefficiency there remains an intriguing open question. Finally, a natural next step is to move from diagnosis to design. Having identified the behavioral sources of inefficiency, future research can test targeted interventions—such as belief-calibration tools, exploration-guidance interfaces, or human–AI collaboration protocols—designed to correct the specific margins on which humans underperform. More generally, the framework can be extended to compare not only human and algorithmic policies, but any two sequential decision rules, making it useful for studying organizational routines, policy changes, and hybrid human–AI workflows in other high-stakes domains.

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Online Appendices

Appendix A: Additional Details on the Datasets

First, we describe how we process the primary dataset on the trial allocation history of the projects during PSI in Appendix A.1. Then, we present the summary statistics for the data during PSI-1 in Appendix A.2. Finally, we describe the full list of project characteristics we constructed in Appendix A.3.

A.1. Data Cleaning Process

Our primary dataset (Berman et al. 2017) records detailed trial level information for all the projects conducted by 40 participating labs from 2000 to 2015. Specifically, for each trial entry to a project, it captures the start and end date of the trials, as well as the dates for the key intermediate steps. The raw data consists of 954,481 trial entries to 335,771 projects. We first removed the projects that have missing protein sequence information or are unable to be matched with their UniProt (UniProt Consortium 2021) entries, and after this step we have 952,107 trial entries to 332,293 projects. Next, we dropped the project and trial entries that were recorded in the data but no actual wet-lab trial effort was allocated, and this step left us with 820,278 trials[1]. Then, we dropped the projects that were terminated not because of failures but extremely high similarity with existing successful projects (i.e., likely duplications), and after this step we have 816,873 trials to 231,805 projects. Finally, there is a small number of projects attempted by scientists across phases (e.g., they might first attempt a project in PSI-1 and then continue working on it in PSI-2). Because laboratory objectives may have shifted between PSI-1 and PSI-2, those projects might introduce additional complication to our analysis. As a result, we dropped the projects that the scientists attempted across phases. This step left us with 709,079 trials to 225,149 projects.

[1] For each attempted trial, the intermediate steps include "cloned", "expressed", "purified", "crystallized/NMR/cryo-EM", and the final step is "deposited". A trial is successful if all the steps succeed sequentially. Note that each intermediate status indicates the current action instead of a successful step (e.g., "cloned" simply means cloning was conducted, and "expressed" means cloning was successful) (Mizianty and Kurgan 2011). Because no recorded wet-lab activity occurs before the "cloned" step, we treat any trial entries that at most never attained the "cloned" status as unattempted and exclude them from our analysis.

A.2. Summary Statistics of the Primary Dataset in PSI-1

Table EC.1 presents the summary statistics for the projects and trials attempted by the scientists during PSI-1.

Table EC.1 Summary Statistics of the Primary Dataset in PSI-1

Definition	JCSG	MCSG	NESG	Others	Total
Project-Level Statistics					
Projects attempted	3,800	12,844	5,058	39,149	60,851
Projects succeeded	230	585	288	968	2,071
Project success rate (%)	6.053	4.555	5.694	2.473	3.403
Average number of trials per project	12.79	1.121	2.977	1.214	2.064
Average number of trials to failed projects	9.086	1.116	2.800	1.213	1.800
Average number of trials to succeeded projects	70.30	1.228	5.899	1.229	9.549
Average number of failed trials before success	61.14	1.099	5.007	1.122	8.322
Trial-Level Statistics					
Trials allocated	48,607	14,396	15,056	47,508	125,567
Trials to successful projects	16,169	718	1,699	1,190	19,776
Trials succeeded	260	660	337	1,031	2,288
Fraction of trials to successful projects (%)	33.26	4.987	11.28	2.505	15.75
Trial success rate (%)	0.535	4.585	2.224	2.170	1.822
Average trial duration in days	571.1	537.4	2,108	302.1	649.8

A.3. List of Project Characteristics

We compile a comprehensive list of project characteristics primarily motivated by scientific reasoning, supplemented with additional contextual features. Table EC.2 describes these variables and their sources in detail.

Table EC.2: Description of Project Characteristics

Variable	Description
$[AAIndexMean]_i$	Attributes of amino acids from the AAindex database (Kawashima et al. 2007). Following Babnigg and Joachimiak (2010), we apply affinity propagation clustering in scikit-learn to obtain 30 clusters. Then, for each attribute of the cluster center, we calculate the local average value for project i using a sliding window with a size of 7 amino acids. This resulted in 30 variables.
$[AAIndexMax]_i$	Attributes of amino acids from the AAindex database (Kawashima et al. 2007). Following Babnigg and Joachimiak (2010), we apply affinity propagation clustering in scikit-learn to obtain 30 clusters. Then, for each attribute of the cluster center, we calculate the local maximum value for project i using a sliding window with a size of 7 amino acids. This resulted in 30 variables.
$[AAIndexMin]_i$	Attributes of amino acids from the AAindex database (Kawashima et al. 2007). Following Babnigg and Joachimiak (2010), we apply affinity propagation clustering in scikit-learn to obtain 30 clusters. Then, for each attribute of the cluster center, we calculate the local minimum value for project i using a sliding window with a size of 7 amino acids. This resulted in 30 variables.
$[aminoAcid]_i$	The number of each amino acid in project i, which is calculated using the Bio.SeqUtils.ProtParam module in Python. Certain amino acids may lead to higher likelihood of success in determining structures (e.g., Price et al. 2009, Babnigg and Joachimiak 2010, Jahandideh et al. 2014). This resulted in 20 variables.
$[aminoAcidPercent]_i$	The percentage of each amino acid in project i, which is calculated using the Bio.SeqUtils.ProtParam module in Python. Certain amino acids may lead to higher likelihood of success in determining structures (e.g., Price et al. 2009, Babnigg and Joachimiak 2010, Jahandideh et al. 2014). This resulted in 20 variables.

Variable	Description
$[embedding]_i$	A 1024-dimensional embedding obtained from the ProtT5-XL-UniRef50[2] model proposed by Elnaggar et al. (2022), which is based on the T5 model (Raffel et al. 2020). For each protein sequence in our data set, we directly apply the model to obtain the embedding without any fine-tuning. This resulted in 1,024 variables.
$[exposedAminoAcid]_i$	The number of each amino acid on the predicted exposed surface for project i, which is calculated using NetSurfP (Klausen et al. 2019). We impose a minimum cut-off value of 0.25 for relative solvent accessibility. Certain amino acids on the exposed surface may lead to higher likelihood of success in determining structures (e.g., Price et al. 2009, Babnigg and Joachimiak 2010, Jahandideh et al. 2014). This resulted in 20 variables.
$[exposedAminoAcidPercent]_i$	The percentage of each amino acid on the predicted exposed surface for project i, which is calculated using NetSurfP (Klausen et al. 2019). We impose a minimum cut-off value of 0.25 for relative solvent accessibility. Certain amino acids on the exposed surface may lead to higher likelihood of success in determining structures (e.g., Price et al. 2009, Babnigg and Joachimiak 2010, Jahandideh et al. 2014). This resulted in 20 variables.
$[gene]_i$	A set of binary indicators for whether project i is coded by certain genes. We collect data from UniProt (UniProt Consortium 2021) and only include genes that (1) have more than 200 occurrences and (2) are from proteins whose structures are successfully published in the data. This resulted in 8 variables.
$[keyword]_i$	A set of binary indicators for whether project i is associated with certain keywords. We collect data from UniProt (UniProt Consortium 2021) and only include keywords that (1) have more than 200 occurrences and (2) are from proteins whose structures are successfully published in the data. This resulted in 352 variables.
$[lab]_i$	A set of binary indicators for the lab of project i. This resulted in 40 variables.
$[phylum]_i$	A set of binary indicators for whether project i is associated with the specific superkingdom and phylum. We collect data from UniProt (UniProt Consortium 2021). This resulted in 109 variables.

[2] See https://huggingface.co/Rostlab/prot_t5_xl_uniref50.

Variable	Description
$biomedical_i$	Binary indicator for whether project i is biomedically important. Following Zhuo (2025), we consider project i biomedically important if in the data the target category list or target rationale[3] it contains the following keywords: activator, adhesion, antibiotic, binding, biochemistry, biological interest, biomedical, cascade, catalyze, cell development, community nominated, community-nominated, community request, conserved, disease, coronavirus, drug, drug development, drug target, effector, enzyme, essential, function, functional studies, functional, gpcr, high value, high-value, hiv, homeostasis, host, immune, immunity, infection, infectious, inhibitor, interaction, interact, legionella, medical school, metabolism, mitochondria, model system, operon, parkinsons, partnership, pathogen, pathology, pathway, phosphatase, pneumonia, protein family of high biological importance, reagent, receptor, resistance, resistant, salmonella, school of medicine, secret, sensor, shen lab, shen_lab, shen selection, stem cell, substrate, syndrome, synthesis, t-cell, tcell, therapeutic, thorson lab, toxoplasma, transcription, transport, tuberculosis, tumor, university, vaccine, vibrio, virulence, virulent. In the comparison with expanded objective considerations, we consider project i to be biomedically important if $biomedical_i = 1$.
$eukaryote_i$	The maximum percentage of identity parts of project i to any eukaryotic molecule. We use DIAMOND (Buchfink et al. 2021) to search the sequences against the UniProt database (UniProt Consortium 2021). In the comparison with expanded objective considerations, we consider project i to be eukaryotic protein if $eukaryote_i = 100$.
$extinctCoeffReduced_i$	The molar extinction coefficient with reduced cysteines of project i, which is calculated using the Bio.SeqUtils.ProtParam module in Python. This feature was used in Slabinski et al. (2007).
$extinctCoeffOxidized_i$	The molar extinction coefficient with disulfid bridges of project i, which is calculated using the Bio.SeqUtils.ProtParam module in Python. This feature was used in Slabinski et al. (2007).
$gravyIndex_i$	Grand average of hydropathicity index of project i, which is calculated using the Bio.SeqUtils.ProtParam module in Python. This feature was used in Slabinski et al. (2007), Price et al. (2009), Jahandideh et al. (2014) and Babnigg and Joachimiak (2010).

[3] In our primary dataset, for each project, it contains target category list and target rationale, which describes the category of the project and rationale of selecting it for structure determination.

Variable	Description
$gaps_i$	The average number of insertions in the alignment for project i relative to the homologs. We use DIAMOND (Buchfink et al. 2021) to search the sequences against the UniProt database (UniProt Consortium 2021). This feature was used in Slabinski et al. (2007), Jaroszewski et al. (2008), and Jahandideh et al. (2014).
$gapOpen_i$	The average number of insertion openings in the alignment for project i relative to the homologs. We use DIAMOND (Buchfink et al. 2021) to search the sequences against the UniProt database (UniProt Consortium 2021). This feature was used in Slabinski et al. (2007), Jaroszewski et al. (2008), and Jahandideh et al. (2014).
$human_i$	The maximum percentage of identity parts of project i to any human molecule. We use DIAMOND (Buchfink et al. 2021) to search the sequences against the UniProt database (UniProt Consortium 2021). In the comparison with expanded objective considerations, we consider project i to be human protein if $human_i = 100$.
$instabilityIndex_i$	Instability index of project i, which is calculated using the Bio.SeqUtils.ProtParam module in Python. This feature was used in Slabinski et al. (2007), Jaroszewski et al. (2008), and Jahandideh et al. (2014).
$isoelectricPoint_i$	Isoelectric point of project i, which is calculated using the Bio.SeqUtils.ProtParam module in Python. This feature was used in Slabinski et al. (2007), Jaroszewski et al. (2008), Price et al. (2009), and Jahandideh et al. (2014).
$membrane_i$	Binary indicator for whether the UniProt information for project i contains the word "membrane." In the comparison with expanded objective considerations, we consider project i to be membrane protein if $membrane_i = 1$.
$molecularWeight_i$	Isoelectric point of project i, which is calculated using the Bio.SeqUtils.ProtParam module in Python.
$novel_i$	Binary indicator for whether project i is novel. Following Zhuo (2025), we consider project i novel if in the data the target category list or target rationale contain the following keywords: big, coverage of protein universe, diversity, first structure of class, low sequence identity, mega, metagenomic, new fold, no structural information, no structure, number of homologs, pfam, remote homologs, structural coverage, structural template for unsolved, structure coverage, unsolved families, without any solved structures, without structure. In the comparison with expanded objective considerations, we consider project i to be novel if $novel_i = 1$.

Variable	Description
$percentCoil_i$	The predicted percentage of coil secondary structure of project i, which is calculated using NetSurfP (Klausen et al. 2019). This feature was used in Slabinski et al. (2007), Price et al. (2009), and Jahandideh et al. (2014).
$percentCoiledCoil_i$	The percentage of coiled-coil regions for project i, which is obtained from UniProt (UniProt Consortium 2021). This feature was used in Slabinski et al. (2007), Jaroszewski et al. (2008), Price et al. (2009), Babnigg and Joachimiak (2010), and Jahandideh et al. (2014).
$percentDisordered_i$	The predicted percentage of disordered region for project i, which is calculated using NetSurfP (Klausen et al. 2019). This feature was used in Slabinski et al. (2007), Jaroszewski et al. (2008), Price et al. (2009), and Jahandideh et al. (2014).
$percentDisorderedUniprot_i$	The percentage of disordered region for project i, which is obtained from UniProt (UniProt Consortium 2021). This feature was used in Slabinski et al. (2007), Jaroszewski et al. (2008), Price et al. (2009), Babnigg and Joachimiak (2010), and Jahandideh et al. (2014).
$percentExposed_i$	The predicted percentage of amino acids on the exposed surface for project i, which is calculated using NetSurfP (Klausen et al. 2019). We impose a minimum cut-off value of 0.25 for relative solvent accessibility. This feature was used in Price et al. (2009), Babnigg and Joachimiak (2010), and Jahandideh et al. (2014).
$percentHelix_i$	The predicted percentage of helix secondary structure for project i, which is calculated using NetSurfP (Klausen et al. 2019). This feature was used in Slabinski et al. (2007), Jaroszewski et al. (2008), Price et al. (2009), Babnigg and Joachimiak (2010), and Jahandideh et al. (2014).
$percentLowComplexity_i$	The predicted percentage of low-complexity regions for project i, which is calculated using SEG (Wootton 1994). This feature was used in Slabinski et al. (2007) and Jaroszewski et al. (2008).
$percentSignalPeptide_i$	The percentage of signal peptide for project i, which is obtained from UniProt (UniProt Consortium 2021). This feature was used in Slabinski et al. (2007), Price et al. (2009), Babnigg and Joachimiak (2010), and Jahandideh et al. (2014).
$percentStrand_i$	The predicted percentage of strand secondary structure for project i, which is calculated using NetSurfP (Klausen et al. 2019). This feature was used inSlabinski et al. (2007), Jaroszewski et al. (2008), Price et al. (2009), Babnigg and Joachimiak (2010), and Jahandideh et al. (2014).

Variable	Description
$percentTransmembraneHelices_i$	The percentage of transmembrane helices for project i, which is obtained from UniProt (UniProt Consortium 2021). This feature was used in Slabinski et al. (2007), Jaroszewski et al. (2008), Price et al. (2009), Babnigg and Joachimiak (2010), and Jahandideh et al. (2014).
$prevPub_i$	The number of publications on project i before the year it was attempted.
$prevStruct_i$	The number of already determined structures in the same protein family with project i before the year it was attempted. We obtain protein family information from UniProt (UniProt Consortium 2021), and match the protein families with structures using EMBL-EBI (EMBL-EBI 2021).
$surfaceRuggedness_i$	The surface ruggedness of project i, which was calculated using NetSurfP (Klausen et al. 2019). This feature was used in Jahandideh et al. (2014).

Appendix B: Additional Details on the Algorithm Setup

Following Hwang et al. (2023), the regret of the MAB model is bounded if the upper confidence bound $u_{\mathcal{F}_t}(q_{ijt}, \boldsymbol{X}_{ij})$ at time t for project j is in the form:

$$u_{\mathcal{F}_t}(q_{ijt}, \boldsymbol{X}_{ij}) = f_{ijt} + \gamma_i \sqrt{\tilde{\boldsymbol{g}}_{ijt}^T \boldsymbol{Z}_{it}^{-1} \tilde{\boldsymbol{g}}_{ijt}/m}, \tag{EC.1}$$

in which f_{ijt} is the predicted reward of allocating q_{ijt} trials by the neural network, $\tilde{\boldsymbol{g}}_{ijt} = \frac{\partial f_{ijt}}{\partial \boldsymbol{\theta}_{it}}$, m is the network width, and $\boldsymbol{Z}_{it}$ is the accumulated design matrix.

In our setting, the neural network predicts the logit for the success rate of one trial on project j (i.e., $y_{ijt} = \mathrm{NN}(\boldsymbol{\theta}_{it}, \boldsymbol{X}_{ij})$), which leads to a success probability $p_{i,\mathcal{F}_t}(\boldsymbol{X}_{ij}) = \sigma(y_{ijt})$, where $\sigma(\cdot)$ is the sigmoid function. Let $g(\boldsymbol{X}_{ij}, \boldsymbol{\theta}_{it}) = \frac{\partial y_{ijt}}{\partial \boldsymbol{\theta}_{it}}$, the derivative of y_{ijt} with respect to parameters in the neural network.

When $\tilde{q}_{ijt} + q_{ijt}$ independent trials are allocated to project j, the predicted reward in Equation (EC.1)—that is, the probability of achieving at least one success—is given by:

$$f_{ijt} = 1 - (1 - p_{i,\mathcal{F}_t}(\boldsymbol{X}_{ij}))^{\tilde{q}_{ijt} + q_{ijt}}.$$

Note that f_{ijt} is an increasing function of $p_{i,\mathcal{F}_t}(\boldsymbol{X}_{ij})$ and therefore of y_{ijt}.

Exploration term via the chain rule. Because $f \circ \sigma$ is monotonically increasing in y, to obtain the exploration value term $B(\boldsymbol{\theta}_{it}, \boldsymbol{X}_{ij}, q_{ijt}) = \sqrt{\boldsymbol{G}_{ijt}^T \boldsymbol{Z}_{it}^{-1} \boldsymbol{G}_{ijt}/m}$ that aligns with the the immediate success term, we can apply the chain rule to obtain:

$$\begin{aligned}
\frac{\partial f_{ijt}}{\partial \boldsymbol{\theta}_{it}} &= \frac{\partial [1 - (1 - p_{i,\mathcal{F}_t}(\boldsymbol{X}_{ij}))^{\tilde{q}_{ijt} + q_{ijt}}]}{\partial \boldsymbol{\theta}_{it}} = \frac{\partial [1 - (1 - p_{i,\mathcal{F}_t}(\boldsymbol{X}_{ij}))^{\tilde{q}_{ijt} + q_{ijt}}]}{\partial p_{i,\mathcal{F}_t}(\boldsymbol{X}_{ij})} \cdot \frac{\partial p_{i,\mathcal{F}_t}(\boldsymbol{X}_{ij})}{\partial y_{ijt}} \cdot \frac{\partial y_{ijt}}{\partial \boldsymbol{\theta}_{it}} \\
&= \frac{\partial [1 - (1 - p_{i,\mathcal{F}_t}(\boldsymbol{X}_{ij}))^{\tilde{q}_{ijt} + q_{ijt}}]}{\partial p_{i,\mathcal{F}_t}(\boldsymbol{X}_{ij})} \cdot \frac{\partial p_{i,\mathcal{F}_t}(\boldsymbol{X}_{ij})}{\partial y_{ijt}} \cdot g(\boldsymbol{X}_{ij}, \boldsymbol{\theta}_{it}) \\
&= (\tilde{q}_{ijt} + q_{ijt})(1 - p_{i,\mathcal{F}_t}(\boldsymbol{X}_{ij}))^{\tilde{q}_{ijt} + q_{ijt} - 1} \cdot \frac{\partial p_{i,\mathcal{F}_t}(\boldsymbol{X}_{ij})}{\partial y_{ijt}} \cdot g(\boldsymbol{X}_{ij}, \boldsymbol{\theta}_{it}).
\end{aligned}$$

Following existing literature on logistic bandits, where the reward is a sigmoid function of the context and action (e.g., Faury et al. 2020), we bound $\frac{\partial p_{i,\mathcal{F}_t}(\boldsymbol{X}_{ij})}{\partial y_{ijt}}$ from below with a constant k_i, and leverage this bound to obtain $\boldsymbol{G}_{ijt}$:

$$\boldsymbol{G}_{ijt}^T = (\tilde{q}_{ijt} + q_{ijt})(1 - p_{i,\mathcal{F}_t}(\boldsymbol{X}_{ij}))^{\tilde{q}_{ijt} + q_{ijt} - 1} \cdot k_i \cdot g(\boldsymbol{X}_{ij}, \boldsymbol{\theta}_{it}).$$

Since k_i is a constant, we can further merge it into γ_i in Equation (EC.1). Therefore, we have

$$B(\boldsymbol{\theta}_{it}, \boldsymbol{X}_{ij}, q_{ijt}) = (\tilde{q}_{ijt} + q_{ijt})(1 - p_{i,\mathcal{F}_t}(\boldsymbol{X}_{ij}))^{\tilde{q}_{ijt} + q_{ijt} - 1} ||g(\boldsymbol{X}_{ij}, \boldsymbol{\theta}_{it})/\sqrt{m}||_{\boldsymbol{Z}_{it}^{-1}},$$

in which $||g(\boldsymbol{X}_{ij}, \boldsymbol{\theta}_{it})/\sqrt{m}||_{\boldsymbol{Z}_{it}^{-1}}$ is the Mahalanobis norm of $g(\boldsymbol{X}_{ij}, \boldsymbol{\theta}_{it})/\sqrt{m}$ with respect to $\boldsymbol{Z}_{it}$.

We also include a implementation note. Consistent with the experimental setup in Hwang et al. (2023) and common practice in neural-bandit studies (e.g., NeuralUCB by Zhou et al. 2020 and NeuralTS by Zhang et al. 2021), we use a constant γ term and omit the offset e_t for simplicity. This choice simplifies implementation without affecting the overall structure of the upper confidence bound formulation.

Appendix C: Additional Details on the Deep Neural Network in Benchmark ML Model

In Appendix C.1, we describe details on the architecture and training procedure of the deep neural network used in our benchmark ML model. In Appendix C.2, we show that the predictions of this neural network align well with the empirical success probabilities.

C.1. Model Architecture and Training Process

The structure of our neural network is shown in Figure EC.1. Specifically, $B = 4096$ (chosen through a grid search of values from $\{512, 1024, 2048, 4096\}$) is the batch size and $Dim = 1729$ (705 biophysical properties and 1024-dim embeddings) is the dimension of inputs. The output of the network is logits, to which we further apply the sigmoid function to transform logits into the predicted probabilities. We use BCEWithLogitsLoss as the loss function, and Adam with a weight decay of 0.00001 (chosen through a grid search of values from $\{0.01, 0.001, 0.0001, 0.00001, 0.000001\}$) as the optimizer. In each period, we retrain the network using the updated training dataset for 30 epochs.

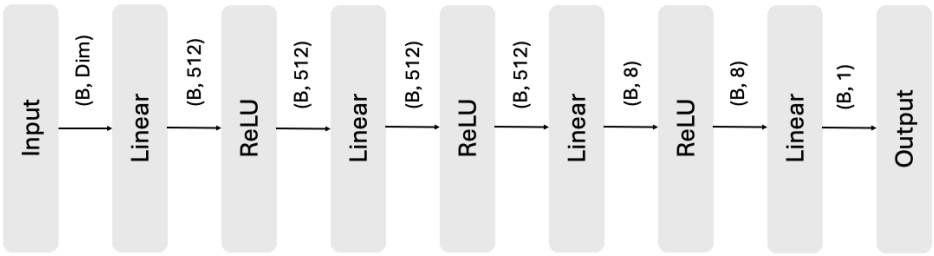


Figure EC.1 Neural Network Structure

Note: Figure EC.1 presents the structure of our neural network. It has three hidden layers, with a batch size of 4,096. The input dimension is 1,729. The hidden layers have a width of 512, 512, and 8 respectively, each followed by a ReLU activation layer.

Directly using the predicted probabilities obtained from classifiers could be problematic. Many highly accurate classification models tend to be miscalibrated, that the predicted probabilities do not align well with the true ones (Naeini and Cooper 2016, Kull et al. 2017, Guo et al. 2017). Therefore, instead of directly using the prediction outcomes by our model, we adopt Platt scaling (Platt 1999), which has been widely adopted and proven effective (Guo et al. 2017), to calibrate our model predictions. Specifically, given the logit output of an classifier $f(x)$, Platt scaling produces the following probability estimate:

$$P(y = 1|x) = \frac{1}{1 + \exp^{-(Af(x)+B)}},$$

in which A and B are parameters to be trained on the held-out calibration set. The training objective is to minimize the Cross Entropy Loss and we train the model for 30 epochs. After obtaining the estimates $\hat{A}$ and $\hat{B}$, the predicted probability after calibration can be directly calculated with $\frac{1}{1+\exp^{-(\hat{A}f(x)+\hat{B})}}$.

In Appendix C.2. where we demonstrate the prediction accuracy of our algorithm, we apply Platt scaling on the validation set. In Section 5 and Section 6 where we compare the dynamic decision making between

machine and human, in each period, we first split 80% of the data to train the model, and apply Platt scaling to the rest 20% data. However, we need to use all the available data up to period t to train the algorithm to obtain predictions, instead of using the model trained using only 80% of data. As a result, after obtaining the estimates for A and B, we retrain the model using all available data up to period t, and use the estimated A and B to calibrate the model predictions obtained with the model trained on the full training data.

C.2. Performance of the ML Model

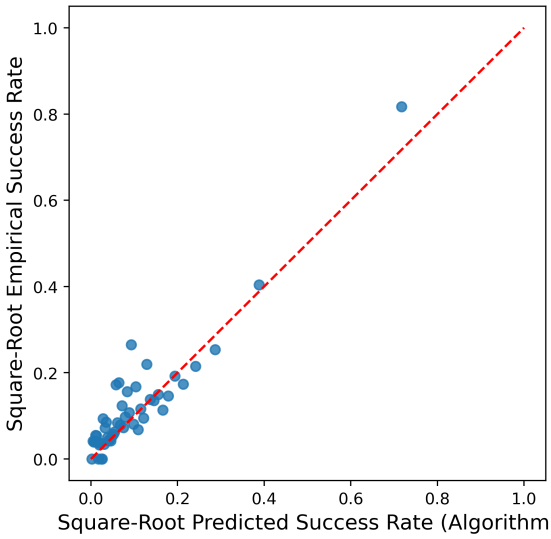


Figure EC.2 Machine Predictions vs. Empirical Success Rate

Note: Figure EC.2 presents the average empirical success rate and the predicted success rate by our ML algorithm. We group the projects into 50 quantile bins based on the predicted success rate, and take the square root of the predicted and empirical success rates.

We evaluate how our model's predicted success probabilities compare to the empirical project outcomes. Specifically, we randomly assign 70% of projects and their associated trials from the whole data (across PSI-1 and PSI-2) to the training set, 15% to a validation set for hyperparameter tuning and model calibration, and the remaining 15% to a held-out test set for model performance evaluation. For each project j in the test set, the model predicts $p^{ML}_{j,\mathcal{F}_t}$, the success probability per trial. We then group projects in the test set into 50 quantile bins based on their predicted probabilities $p^{ML}_{j,\mathcal{F}_t}$, and calculate the average predicted probability within each bin. Because for most projects the success rates are very small, we take the square root of the predicted and empirical success rates for better visualization. Figure EC.2 shows the average empirical success rate against the mean predicted probability for each bin. The data points are centered around the diagonal line, suggesting that the machine prediction aligns well with the empirical probability of success.

Appendix D: Additional Details on Estimating the Human Decision Making Model

We first elaborate how we construct $P(W_{ijt}^k = 1|\boldsymbol{\theta}_{it}^{Ho}, \gamma_i^H, \beta_i^H)$ from $V(W_{ijt}^k = 1|\boldsymbol{\theta}_{it}^{Ho}, \gamma_i^H, \beta_i^H)$. For humans, solving Equation (1) is equivalent to solving

$$\begin{aligned} &\max_{\boldsymbol{q}_{it}} \sum_j \sum_k V(W_{ijt}^k = 1|\boldsymbol{\theta}_{it}^{Ho}, \gamma_i^H, \beta_i^H), \\ &\text{s.t. } \sum_j q_{ijt} \leq Q_{it}, \end{aligned}$$

in which

$$\begin{aligned} V(W_{ijt}^k = 1|\boldsymbol{\theta}_{it}^{Ho}, \gamma_i^H, \beta_i^H) &= u_{\mathcal{F}_t}^H(q_{ijt} = k, \boldsymbol{X}_{ij}) - u_{\mathcal{F}_t}^H(q_{ijt} = k-1, \boldsymbol{X}_{ij}) + \epsilon_{ijt} \\ &= U(W_{ijt}^k = 1|\boldsymbol{\theta}_{it}^{Ho}, \gamma_i^H, \beta_i^H) + \epsilon_{ijkt}. \end{aligned}$$

We introduce the error term ϵ_{ijt} to capture the unobserved factors in human decision making, and assume it follows the i.i.d. Type-1 Extreme Value distribution. In the optimization, humans first calculate the value of $V(W_{ijt}^k = 1|\boldsymbol{\theta}_{it}^{Ho}, \gamma_i^H, \beta_i^H)$ for each potential trial k to project j in period t. Then, humans allocate Q_{it} number of trials with the highest $V(\cdot)$ values.

To circumvent the computational burden of the combinatorial assignment problem, we adopt a separability approximation to obtain the choice probabilities: labs make independent, per-trial decisions by comparing each V_{ijtk} to an endogenously determined threshold value τ_{it}, and a trial is allocated iff $V_{ijtk} > \tau_{it}$. For lab i in period t with trial capacity Q_{it}, the threshold is the Q_{it}-th highest value $\{V(\cdot)\}$ of all possible trials. Without loss of generality, suppose this corresponds to trial v on project l. Then, the threshold value τ_{it} is given as:

$$\tau_{it} := V(W_{ilt}^v = 1|\boldsymbol{\theta}_{it}^{Ho}, \gamma_i^H, \beta_i^H) = U(W_{ilt}^v = 1|\boldsymbol{\theta}_{it}^{Ho}, \gamma_i^H, \beta_i^H) + \epsilon_{ilvt},$$

And the probability of allocating trial k to project j in period t can be written as:

$$\begin{aligned} P(W_{ijt}^k = 1|\boldsymbol{\theta}_{it}^{Ho}, \gamma_i^H, \beta_i^H) &= P\left(V(W_{ijt}^k = 1|\boldsymbol{\theta}_{it}^{Ho}, \gamma_i^H, \beta_i^H) > V(W_{ilt}^v = 1|\boldsymbol{\theta}_{it}^{Ho}, \gamma_i^H, \beta_i^H)\right) \\ &= \frac{\exp\{U(W_{ijt}^k = 1|\boldsymbol{\theta}_{it}^{Ho}, \gamma_i^H, \beta_i^H)\}}{\exp\{U(W_{ijt}^k = 1|\boldsymbol{\theta}_{it}^{Ho}, \gamma_i^H, \beta_i^H)\} + \exp\{U(W_{ilt}^v = 1|\boldsymbol{\theta}_{it}^{Ho}, \gamma_i^H, \beta_i^H)\}}. \end{aligned}$$

The choice set of the labs consists of projects they had attempted in the past and those they would attempt within six months following period t. Note that the number of possible trials to allocate could be extremely huge in this case, given that we have a large number of projects and trials to allocate. Therefore, it is impossible to include all the potential trial allocations into our model. To reduce the size of choice sets in our estimation, we take subsamples from the entire set of potential allocations following existing literature on discrete choice analysis with a large number of alternatives (e.g., McFadden 1977, Bierlaire and Krueger 2024, Zhuo 2025). Specifically, we keep all the trials that were empirically allocated by humans. For the trials not allocated by humans, we randomly sample an unallocated trial for each project in the focal period. See Table EC.3 for details.

Table EC.3 Trials Included for Estimation

Project	Trial	Allocated in Period t ?	Notes
j	$\tilde{q}_{ijt}+1$	Y	In period t, the trial capacity for lab i is Q_{it}. k trials are empirically allocated to project j, and $\tilde{q}_{ijt}$ trials are pending in period t. All the k allocated trials in period t are included (from trial $\tilde{q}_{ijt}+1$ to trial $\tilde{q}_{ijt}+k$). For the not allocated trials (from trial $\tilde{q}_{ijt}+k+1$ to trial $\tilde{q}_{ijt}+Q_{it}$), we randomly select one trial (in this example, trial $\tilde{q}_{ijt}+m$) and include it in our estimation.
	...	...	
	$\tilde{q}_{ijt}+k$	Y	
	$\tilde{q}_{ijt}+k+1$	N	
	...	...	
	$\tilde{q}_{ijt}+m-1$	N	
	$\tilde{q}_{ijt}+m$	N	
	...	...	
	$\tilde{q}_{ijt}+Q_{it}$	N	

For identification purposes, we fix the bias term in the output layer of the "Human Network" with that of the auxiliary network. We also replace $g(\boldsymbol{X}_{ij}, \boldsymbol{\theta}_{it}^{H})$ with the counterpart obtained from the auxiliary neural network (i.e., $g(\boldsymbol{X}_{ij}, \boldsymbol{\theta}_{it}^{A})$) in our estimation. Our estimation procedure is as follows: (1) Fix $\boldsymbol{\theta}_{it}^{Ho}$, and estimate γ_i^H, β_i^H. (2) Fix γ_i^H, β_i^H, and estimate $\boldsymbol{\theta}_{it}^{Ho}$. (3) Repeat Steps 1-2 until convergence is reached.

Because the choice set for each lab was evolving over time, and the number of trials allocated in each period was highly volatile (in some periods, only a few dozens of trials were allocated, while in some other periods tens of thousands of trials were allocated), it is possible that during some periods, the projects attempted by the labs had too few variations in characteristics to identify the output layer weights of the "Human Network". Thus, instead of estimating the output layer weights for every single period, we allow the them to be shared across every 10 periods. For example, during periods 0-9, in Step 1 of Figure 3, we fix the output layer weights of the auxiliary neural network with $\boldsymbol{\theta}_{i1}^{Ao}$, the weights obtained at the beginning of period $t=0$. Then in Step 2, we estimate $\boldsymbol{\theta}_{it}^{Ho}$ with the empirical trial allocation decisions of humans during period 0-9. In this case, during periods 0-9, the "Human Network" share the same output layer weights. Note that even though we only update the output layer weights every 10 periods, it does not mean that human predictions for the projects also remain unchanged for 10 periods, because the rest of the parameters in the "Human Network" are still updated every period.

Despite sharing output-layer weights across 10 periods, some last-layer weights of the "Human Network" remain not point-identified. Further expanding the sharing window to 20–30 periods would provide more statistical leverage but would impose an overly restrictive assumption of unlikely low-frequency human updating. We therefore treat these weights as high-dimensional nuisance parameters and do not pursue perfect point identification (Wei and Jiang 2025). Importantly, Figure 4 shows that even with a small set of these nuisance parameters not point identified, the simulated outcomes still match well with the empirical data. Tables EC.4-EC.6 present the estimation results for the three focal labs.

Table EC.4 Estimation Results of "Human Network" Output Layer Weights for JCSG

Periods	Estimated Output Layer Weights ($\hat{\boldsymbol{\theta}}_{it}^{Ho}$)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
0-9	-0.30698	-0.02248	-0.29445	-0.06530	-0.05293	0.80673	-0.76014	-0.02254
	(0.00009)	(0.00007)	-	(0.00009)	(0.00013)	-	-	(0.00012)
10-19	0.06259	-0.36329	0.16026	0.00287	-0.62935	1.00800	-0.44128	0.11827
	(0.00005)	(0.00007)	-	(0.00006)	(0.00014)	-	(0.00007)	(0.00006)
20-29	-0.42133	0.08341	0.01691	0.64553	0.11404	0.86732	-1.09370	0.20104
	(0.00004)	(0.00001)	-	(0.00004)	(0.00001)	(0.11630)	(0.00006)	(0.00002)
30-39	-0.16357	-0.14146	0.63819	0.08034	-0.07563	0.58168	-0.82202	0.10797
	(0.00004)	(0.00003)	(0.03754)	(0.00004)	(0.00004)	(0.05895)	(0.00004)	(0.00005)
40-49	-0.13763	-0.23064	0.35291	0.02765	0.09674	0.97577	-0.84060	0.22328
	(0.00003)	-	-	-	-	-	-	(0.00007)
50-59	-0.14223	-0.21257	-0.11217	0.39913	-0.37200	-1.09520	-0.73161	-0.06156
	(0.00006)	(0.00007)	(0.05114)	(0.00013)	(0.00006)	(0.04576)	(0.00006)	(0.00005)

Note: Standard deviations in parentheses.

Table EC.5 Estimation Results of "Human Network" Output Layer Weights for MCSG

Periods	Estimated Output Layer Weights ($\hat{\boldsymbol{\theta}}_{it}^{Ho}$)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
0-9	-2.11879	-0.30858	-0.28407	-0.48177	0.70898	0.02683	-1.14651	-0.74077
	(0.00631)	(0.00136)	(0.00008)	(0.12040)	(0.00146)	(0.04583)	(0.00439)	(0.00398)
10-19	-0.29045	-0.72672	1.08315	-0.64129	-0.48382	0.37208	-0.89123	-0.55740
	(0.00010)	(0.00019)	(0.00010)	(0.00023)	(0.00012)	-	(0.00024)	(0.00015)
20-29	-0.77004	-0.57590	0.50594	-0.69116	-0.24699	0.86961	-0.57252	-0.36755
	(0.00059)	(0.00043)	(0.00028)	(0.00026)	(0.00019)	(0.03588)	(0.00039)	(0.00020)
30-39	-0.49520	-0.52005	0.42439	-0.54262	-0.33532	0.63245	-0.85550	-0.10758
	(0.00023)	(0.00021)	(0.00017)	(0.00013)	(0.00017)	(0.00032)	(0.00043)	(0.00006)
40-49	-0.61176	-0.39586	0.62503	-0.78859	-0.20657	0.98037	-0.39710	-0.32680
	(0.00019)	(0.00010)	(0.00019)	(0.00014)	(0.00009)	-	(0.00012)	(0.00011)
50-59	-0.79263	-0.45832	0.63611	-0.75101	-0.19960	0.78944	-0.51299	-0.44284
	(0.00032)	(0.00011)	(0.01158)	(0.00026)	(0.00009)	(0.00036)	(0.00013)	(0.00019)

Note: Standard deviations in parentheses.

Table EC.6 Estimation Results of "Human Network" Output Layer Weights for NESG

Periods	Estimated Output Layer Weights ($\hat{\boldsymbol{\theta}}_{it}^{Ho}$)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
0-9	-0.25154	0.21595	0.07234	-0.40884	0.04924	-5.54265	-0.49021	0.01159
	(0.00013)	(0.00021)	(0.00010)	(0.00167)	(0.00015)	(1.93265)	(0.00012)	(0.00052)
10-19	-0.07038	0.13535	-1.31210	-0.17194	-0.37272	-0.65313	-0.44349	-0.10596
	(0.00034)	(0.00046)	(0.00285)	(0.00062)	(0.00089)	-	(0.00068)	(0.00034)
20-29	0.25986	0.30272	-1.48813	-1.47627	-0.68287	0.86304	-0.53415	0.44917
	(0.00005)	(0.00005)	(0.00166)	(0.00028)	(0.00012)	-	(0.00009)	(0.00008)
30-39	-0.06447	-0.09091	-0.33221	-0.10917	-0.15599	-0.73926	-0.46305	-0.01473
	(0.00028)	(0.00051)	(0.00238)	(0.00045)	(0.00044)	(0.06633)	(0.00038)	(0.00045)
40-49	-0.01994	-0.06757	-0.22514	-0.17875	-0.12594	0.91037	-0.59970	0.28469
	(0.00010)	(0.00013)	(0.00019)	(0.00020)	(0.00010)	-	(0.00009)	(0.00015)
50-59	0.09708	0.08930	-0.22041	-0.17365	-0.14932	-0.27089	-0.65992	-0.07334
	(0.00011)	(0.00018)	(0.00015)	(0.00041)	(0.00028)	(0.00117)	(0.00004)	(0.00007)

Note: Standard deviations in parentheses.

Appendix E: Simulation Procedure and Model Fit of the Estimated Human Decision Making Model

In Appendix E.1, we provide more details on the simulation procedure. Then, we present the additional model-fit evaluation results in Appendix E.2.

E.1. Simulation Procedure

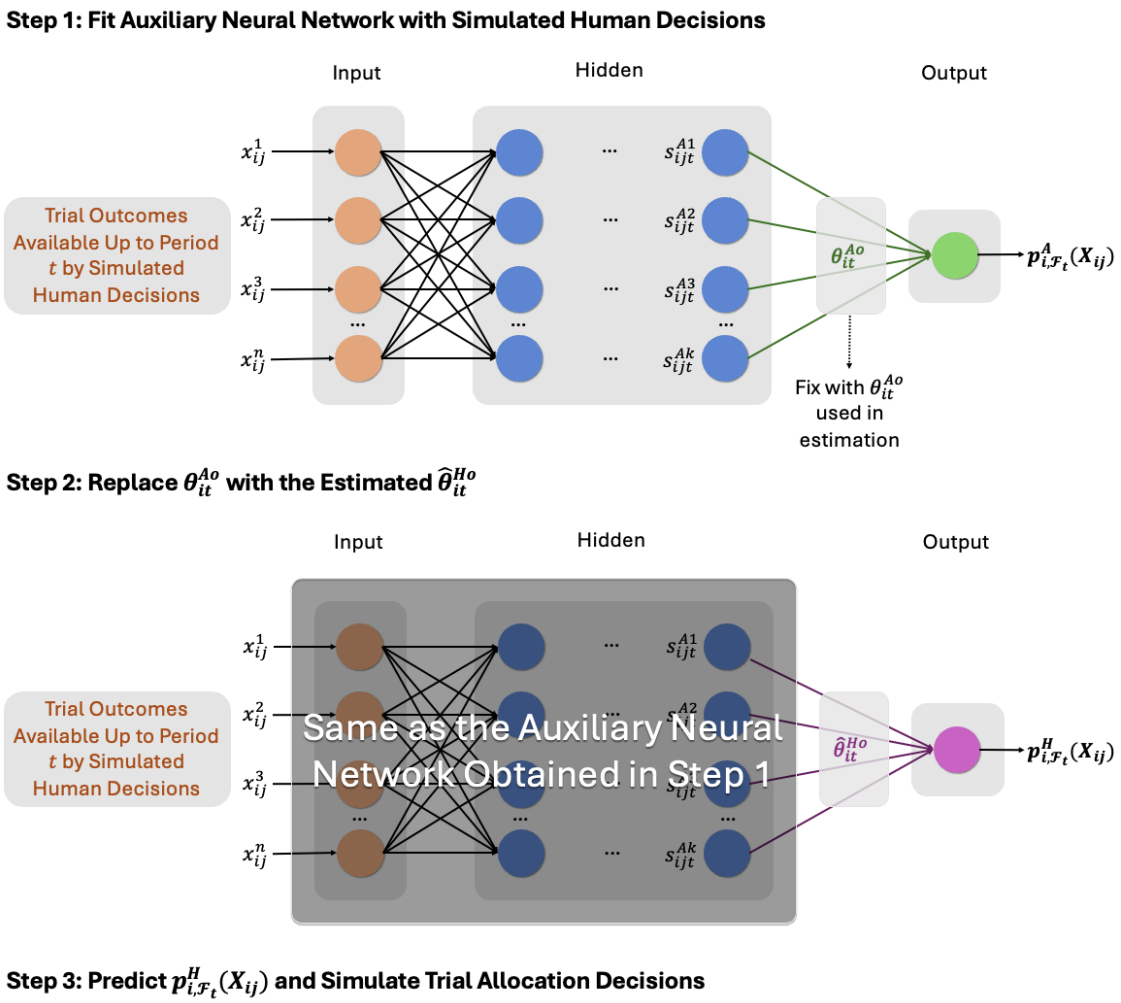


Figure EC.3 Procedure to Obtain Human Predictions in Simulation

Note: Figure EC.3 presents the procedure to obtain human predictions during simulation. We first fit an auxiliary neural network with the data from the simulated human decisions. Note that for consistency, when fitting the auxiliary neural network, we freeze the output layer parameters with $\boldsymbol{\theta}_{it}^{Ao}$, the output layer parameters used during the estimation process in period t. Then, in Step 2, we freeze the input and hidden layers of the auxiliary neural network, and replace the output layer with $\hat{\boldsymbol{\theta}}_{it}^{Ho}$, the estimated output layer parameters for the "Human Network". Finally, we calculate $p_{i,\mathcal{F}_t}^{H}(\boldsymbol{X}_{ij})$ using the neural network obtained in Step 2.

After estimating the parameters, we are able to simulate counterfactual human decisions. The simulation procedure is shown in Figure EC.3. Specifically, in each period t, we first train an auxiliary neural network with the trial outcomes available up to the current period from the simulated human decision-making as

training data. Note that during the training process of the auxiliary neural network in simulation, we freeze the output layer parameters with $\boldsymbol{\theta}_{it}^{Ao}$, the output layer parameters used during the estimation process in period t for consistency. Then, in Step 2, we replace the output layer of the fitted auxiliary neural network $\boldsymbol{\theta}_{it}^{Ao}$ with $\hat{\boldsymbol{\theta}}_{it}^{Ho}$, the estimated output layer parameters for the "Human Network". Finally, in Step 3, we obtain $p_{i,\mathcal{F}_t}^{H}(\boldsymbol{X}_{ij})$, the simulated trial success rate predictions by humans. After obtaining the probabilities, we simulate human trial allocation decisions in this period using $p_{i,\mathcal{F}_t}^{H}(\boldsymbol{X}_{ij})$ and the estimated $\hat{\gamma}_i^H$ and $\hat{\beta}_i^H$. Specifically, in each period t, we allow the simulated human decision maker to have the same trial capacity as actual humans, and simulate the trial allocation decisions. Same as Section 5 and 6, the choice set of the simulated human decision maker consists of projects that humans had attempted in the past and those they would attempt within six periods following period t. In addition, the trial outcome sequence is the same for all the projects as in the empirical data, and for all the trials allocated by our model, their length of duration also stays the same as in the empirical data. Finally, we also impose the restriction that for any project j, the number of trials allocated by the simulated human decision maker $\sum_t q_{ijt}$ should not exceed the total number of trials allocated to project j by lab i empirically in the data for consistency.

E.2. Additional Model Fit Evaluations

In addition to showing that the estimated model replicates overall success in Figure 4, we also examine the number of projects attempted over time. Figure EC.4 plots the cumulative number of unique projects attempted over the 60 periods: the blue line corresponds to the simulated model, and the orange line corresponds to the true empirical trajectory. The two curves remain close to each other, indicating that the estimated structural model accurately replicates human allocation behavior in terms of the number of projects attempted.

Besides, we compare the distribution of trials across projects in each period between empirical and simulated allocations. Specifically, we conduct the Kolmogorov–Smirnov (KS) test to compare the empirical cumulative distributions of trials allocated by scientists and the simulated model. In each period, we compute the KS statistic and its associated p-value under the null hypothesis that the simulated and empirical allocations are from the same empirical cumulative distribution. We plot the p-values over time in Figure EC.5. The red dashed line corresponds to the threshold value of 0.05. The results show that the majority of periods have a p-value greater than 0.05, indicating that the simulated trial allocations do not statistically differ from the empirical human allocations in most periods.

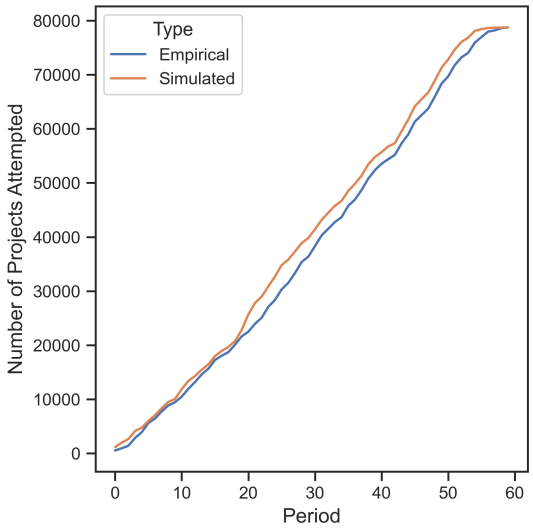


Figure EC.4 Projects Attempted from Simulated and Empirical Allocation

Note: Figure EC.4 presents the cumulative number of unique projects attempted over time from the empirical and simulated allocations. During the simulation process, only project–trial pairs with empirically observed results can be allocated. The choice set in each period is limited to projects that humans had already attempted or would attempt within the next six periods. Trial outcomes and durations replicate the exact empirical sequence.

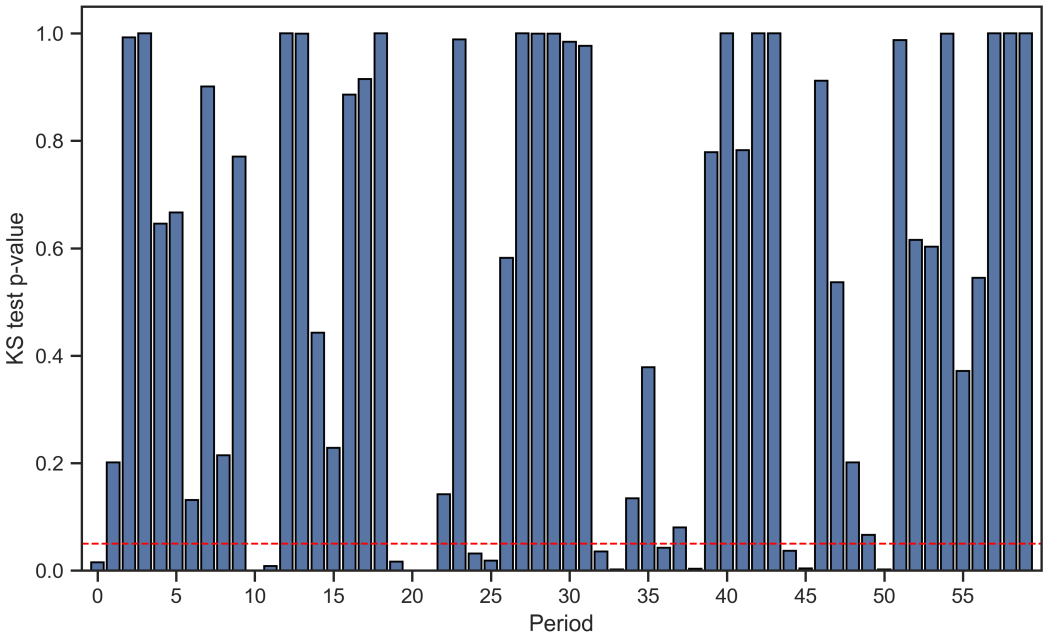
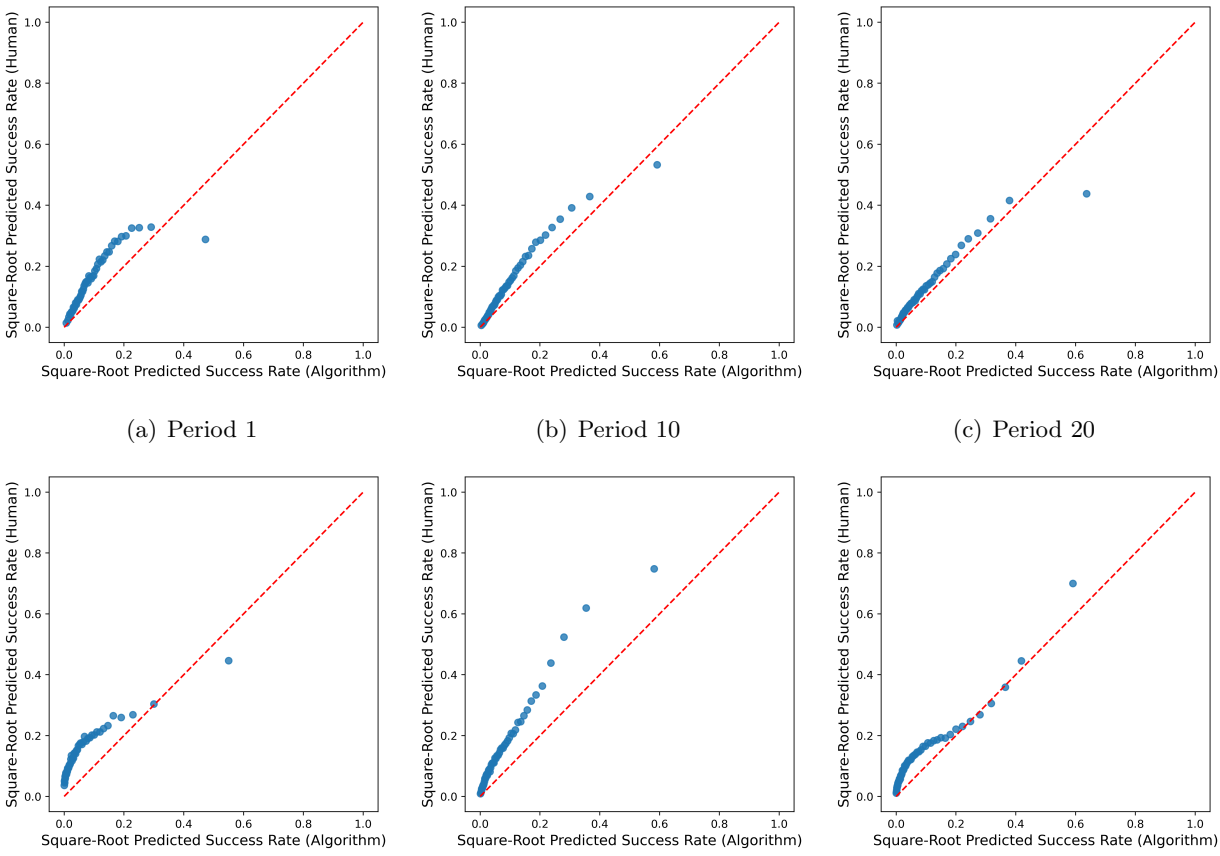


Figure EC.5 Kolmogorov–Smirnov Test Results

Note: Figure EC.5 presents the results of Kolmogorov–Smirnov tests, which compare the empirical cumulative distributions of trials from the empirical and simulated allocations in each period. The x-axis is period, and the y-axis is p-value of the test, with a p-value greater than 0.05 indicating a lack of statistically significant difference between the distribution of the simulated trial allocations and empirical human allocations.

Appendix F: Additional Figures



(a) Period 1 (b) Period 10 (c) Period 20

(d) Period 30 (e) Period 40 (f) Period 50

Figure EC.6 Patterns in Human Prediction Errors Over Time

Note: Figure EC.6 compares the human and algorithm predictions for projects attempted in PSI-2 at different times during PSI-2. Specifically, we train the "Human Network" and our benchmark algorithm leveraging the information set in Period 1, 10, 20, 30, 40, and 50 of PSI-2 obtained from empirical trial allocations respectively. Then, we bin the projects into 50 quantile groups based on the algorithm predicted success rate, and compare against the mean predicted success rate by humans. Because the human and algorithm forecasts are on different scales, we compare them after min–max normalizing each lab's predictions to [0,1].

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